Network Topology — Defense-in-Depth Enterprise Network

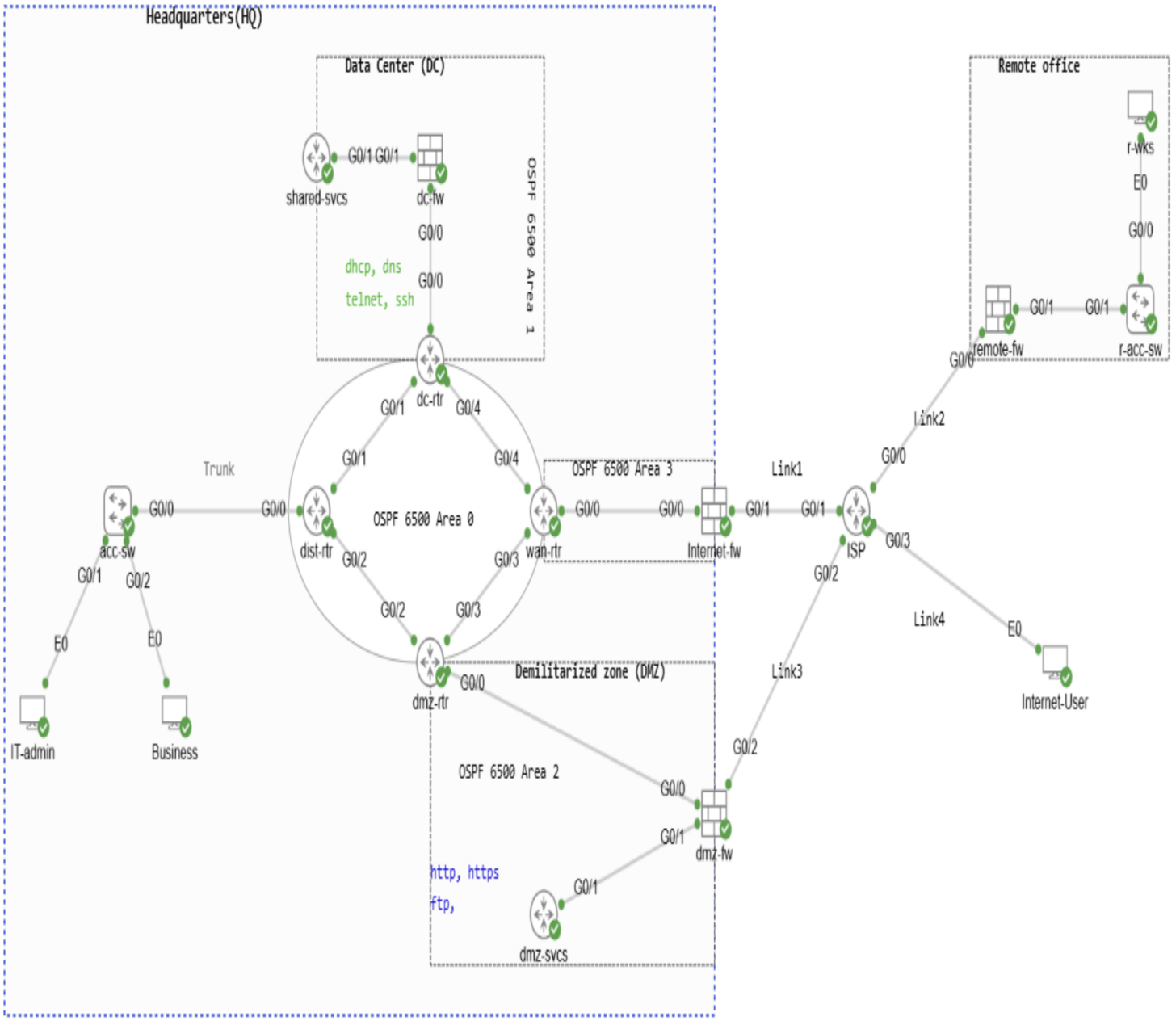


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INTRODUCTION

The goal of this project was to architect a high-performance, secure corporate network that connects a main Headquarters to a Data Center, a DMZ, and a Remote Office. In a modern business environment, it is not enough to just have connectivity; the network must be "invisible" to outside threats while remaining fast and reliable for internal employees. This project focuses on building a "Defense-in-Depth" infrastructure where security is baked into every layer—from the way users get their IP addresses to how offices communicate over the public internet. By the end of this implementation, we have created a network that allows for seamless data flow while strictly hiding our internal structure from the outside world.

Task 1: Configuring IPv4 Addressing and Network Devices.

Objective: By completing this task, students should Design an IP address scheme for the given topology, understanding where to use The private IP addresses or public IP addresses based on the RFC 1918 guidelines. This will enhance their understanding of the practical applications of private and public IP addresses, the importance of proper IP planning, and the distinctions between the infrastructure and endpoint IP address requirements.

"The IP addressing scheme for this topology utilizes a hierarchical design to ensure scalability and security. We implemented **RFC 1918 Private Addressing** for all internal HQ and Data Center segments to conserve public IPv4 space and provide a layer of security through NAT/PAT at the network edge."

Segment	Device	Interface	IP Address	Type	Role
Public / ISP	ISP	G0/1	200.0.0.1	Public	Gateway to Internet User
	ISP	G0/2	200.0.1.1	Public	Gateway to DMZ
	ISP	G0/0	200.0.2.1	Public	Gateway to Remote Office
	ISP	G0/3	200.0.3.1	Public	Internet-User segment
	ISP	Lo0	8.8.8.8	Public	Simulated Global DNS
HQ User LAN	dist-rtr	G0/0.10	192.168.10.1	Private (C)	IT-Admin Gateway (VLAN 10)
	dist-rtr	G0/0.20	192.168.20.1	Private (C)	Business Gateway (VLAN 20)
	acc-sw	Vlan10	192.168.10.2	Private (C)	Mgmt IP - VLAN 10
	acc-sw	Vlan20	192.168.20.2	Private (C)	Mgmt IP - VLAN 20
Data Center	shared-svcs	G0/1	10.1.10.10	Private (A)	DHCP/DNS/Mgmt Server

	dc-fw	G0/1	10.1.10.1	Private (A)	DC Inside Gateway
OSPF Core	dist-rtr	G0/1	10.0.0.1	Private (A)	Link to DC Core
	dc-rtr	G0/1	10.0.0.2	Private (A)	Link to Distribution Core
	dc-rtr	G0/4	10.0.0.5	Private (A)	Link to WAN
	wan-rtr	G0/4	10.0.0.6	Private (A)	Link to DC
	wan-rtr	G0/3	10.0.0.9	Private (A)	Link to DMZ
	dmz-rtr	G0/3	10.0.0.10	Private (A)	Link to WAN
DMZ	dmz-fw	G0/1	10.50.0.1	Private (A)	DMZ Inside Gateway
	dmz-svcs	G0/1	10.50.0.10	Private (A)	Web/Mail Services
Remote Office	remote-fw	G0/1	192.168.100.1	Private (C)	Remote Office Gateway

Task 2: VLANs and Routing Protocol Configuration And Authentication

Objective:By completing this task, students will learn how to segment The network traffic,configure VLANs, establish interconnection between switches and routers, and implement dynamic routing protocols authentication. Routing protocol authentication ensures that only authorized devices can participate in the routing domain, preventing unauthorized or malicious devices from injecting false routing updates. This enhances the network security by protecting against threats such as route hijacking,man-in-the-middle attacks, and accidental misconfigurations.

First We segmented the network into **VLAN 10 (IT)** and **VLAN 20 (Business)** to isolate broadcast domains and improve security. We used **802.1Q trunking** between the access switch and the distribution router to allow for controlled Inter-VLAN routing. This setup ensures that all traffic between departments must pass through a Layer 3 device where security policies can be enforced.By default, all devices on a switch are in the same broadcast domain (VLAN 1). Moving them into separate VLANs ensures that sensitive IT management traffic is isolated from general business traffic.

```
acc-sw#show vlan brief

VLAN Name                                Status    Ports
---- ----------------------------------- --------- -------
1    default                             active    Gi0/3
10   IT-admin                            active    Gi0/1
20   Business                            active    Gi0/2
1002 fddi-default                        act/unsup
1003 token-ring-default                  act/unsup
1004 fddinet-default                     act/unsup
1005 trnet-default                       act/unsup
acc-sw#
```

For a PC to get an IP, the request must climb **up** to the gateway, and the response must climb **down** to the correct PC.

1. The "Request" Path (Going UP)

The PCs don't know where the DHCP server is, so they yell (Broadcast). The **Distribution Router** hears them on its sub-interfaces and acts as a "Relay Agent."

- **IT-Admin PC (VLAN 10):** Uses Gateway **192.168.10.1** ($G0/0.10$).
- **Business PC (VLAN 20):** Uses Gateway **192.168.20.1** ($G0/0.20$).

2. The "Response" Path (Coming DOWN)

The DHCP Server ($10.1.10.10$) sends the IP address back to the **dist-rtr**. The router then has to decide which "door" (Gateway) to use to find the PC.

- **To find the IT-Admin PC:** The router pushes the data out of **G0/0.10**.
- **To find the Business PC:** The router pushes the data out of **G0/0.20**.

3. To verify the Routing Isolation:

To verify the successful segmentation of the network, we performed a routing table check on the IT-DEPT endpoint. The output confirms that the device successfully obtained an IP address (192.168.10.3) and its corresponding default gateway (192.168.10.1) via DHCP. This validates that the 802.1Q trunking between the access switch and distribution router is correctly tagging VLAN 10 traffic and that the router is successfully relaying DHCP requests to the central server.

```
IT-DEPT:~$ ip route
default via 192.168.10.1 dev eth0  metric 202
192.168.10.0/24 dev eth0 scope link  src 192.168.10.3
IT-DEPT:~$
```

```
Business-PC:~$ ip route
default via 192.168.20.1 dev eth0  metric 202
192.168.20.0/24 dev eth0 scope link  src 192.168.20.3
Business-PC:~$
```

4.Verify Message Authentication Implementation.

```
dist-rtr#show ip ospf interface g0/1
GigabitEthernet0/1 is up, line protocol is up
  Internet Address 10.0.0.1/30, Area 0, Attached via Network Statement
  Process ID 6500, Router ID 192.168.20.1, Network Type BROADCAST, Cost: 1
  Topology-MTID    Cost    Disabled    Shutdown      Topology Name
        0           1         no          no            Base
  Transmit Delay is 1 sec, State DR, Priority 1
  Designated Router (ID) 192.168.20.1, Interface address 10.0.0.1
  Backup Designated router (ID) 10.0.0.17, Interface address 10.0.0.2
  Timer intervals configured, Hello 10, Dead 40, Wait 40, Retransmit 5
    oob-resync timeout 40
    Hello due in 00:00:07
  Supports Link-local Signaling (LLS)
  Cisco NSF helper support enabled
  IETF NSF helper support enabled
  Index 1/1/1, flood queue length 0
  Next 0x0(0)/0x0(0)/0x0(0)
  Last flood scan length is 0, maximum is 4
  Last flood scan time is 0 msec, maximum is 1 msec
  Neighbor Count is 1, Adjacent neighbor count is 1
    Adjacent with neighbor 10.0.0.17  (Backup Designated Router)
  Suppress hello for 0 neighbor(s)
  Cryptographic authentication enabled
    Youngest key id is 1
dist-rtr#
```

Dist-rtr

Command Used: dist-rtr# show ip ospf interface g0/1

Key Evidence from Output:

- **Cryptographic authentication enabled**: Confirms the interface is no longer sending clear-text updates.
- **Youngest key id is 1**: Matches the Key ID configured on the neighbor (dc-rtr), ensuring synchronization.
- **Adjacent neighbor count is 1**: This is the ultimate proof. If the MD5 keys did not match, the neighbor relationship would drop to 0, and all routing would fail.

5. The Mechanism:

We utilized a **Message-Digest (MD5)** algorithm for authentication. Unlike simple passwords, MD5 does not send the password over the wire. Instead, it creates a unique 'hash' of the OSPF packet combined with our secret key (cisco123). The receiving router performs the same calculation; if the hashes match, the routing update is accepted. This ensures:

- **Integrity:** The routing update was not changed in transit.
- **Authenticity:** The update came from a trusted pee

If we remove OSPF authentication from interface the neighbor went down immediately.

```
dmz-rtr(config)#int g0/3
dmz-rtr(config-if)#no ip ospf authentication message-digest
dmz-rtr(config-if)#no ip ospf message-digest-key 1 md5 cisco123
```

dmz-rtr

```
wan-rtr#show ip ospf neighbor

Neighbor ID     Pri   State           Dead Time   Address         Interface
10.0.0.17         1   FULL/BDR        00:00:38    10.0.0.5        GigabitEthernet0/4
10.0.0.22         1   FULL/DR         00:00:37    10.0.0.22       GigabitEthernet0/0
```

Wan-rtr

After Again we add the authentication and then router came up immediately.

```
dmz-rtr(config)#int g0/3
dmz-rtr(config-if)# ip ospf message-digest-key 1 md5 cisco123
dmz-rtr(config-if)# ip ospf authentication message-digest
dmz-rtr(config-if)#end
dmz-rtr#
```

dmz-rtr

```
wan-rtr#show ip ospf neighbor

Neighbor ID     Pri   State           Dead Time   Address         Interface
10.0.0.25         1   FULL/DR         00:00:39    10.0.0.10       GigabitEthernet0/3
10.0.0.17         1   FULL/BDR        00:00:39    10.0.0.5        GigabitEthernet0/4
10.0.0.22         1   FULL/DR         00:00:39    10.0.0.22       GigabitEthernet0/0
wan-rtr#
```

Wan-rtr

Task 3: ASA Basic Terminologies and Configuration

Objective: In this task, students will gain hands-on experience in configuring basic settings on Cisco ASA firewalls, including hostnames, interface IP addresses, security zones, security levels, and OSPF routing. The task will also familiarize students with key ASA concepts such as zones, security levels, and stateful inspection. By completing this task, students will understand the fundamental principles of ASA configuration and dynamic routing protocols, as well as the role that ASA plays in traffic segmentation and network protection.

The Functional Logic of the ASA

The Cisco ASA acts as the "intelligent gatekeeper" of the network by organizing interfaces into logical **Security Zones** (such as inside, outside, and dmz), each assigned a numerical **Security Level** from **0** (untrusted) to **100** (most trusted). Unlike a standard router that simply forwards packets, the ASA uses a **Stateful Inspection** engine; this means it monitors the state of active connections and automatically allows return traffic from a lower security level to a higher one if the session was initiated from the inside. By integrating **OSPF dynamic routing** into this secure framework, the ASA ensures that while the network remains reachable and scalable, no unauthorized external traffic can penetrate the private subnets without an explicit access rule.

Key Terminologies

- **Nameif (Interface Naming)** The process of giving a physical port a functional name like inside or outside.
- **Security Levels:** The hierarchy (0–100) that determines the default flow of traffic.
- **Stateful Inspection:** The ability of the firewall to "remember" outgoing requests so it can safely let the replies back in.
- **Security Zones:** The logical segmentation of the network (e.g., separating the DMZ from the Internal LAN).

```
dc-fw# show nameif
Interface                Name                        Security
GigabitEthernet0/0       outside                       0
GigabitEthernet0/1       inside                      100
dc-fw#
```

Dc-fw

The Cisco ASA serves as the network's security perimeter by enforcing traffic flow based on **Security Levels** and **Stateful Inspection**. Unlike traditional routers, the ASA assigns a "trust" value to each interface: the **inside** zone is assigned **Security Level 100** (most trusted), while the **outside** zone is assigned **Security Level 0** (untrusted). By default, the ASA allows traffic to move from a higher security level to a lower one but blocks all unsolicited traffic originating from the outside.

When the **IT-DEPT PC** (192.168.10.X) sends data to the Data Center, the ASA inspects the packet and creates an entry in its state table. This **Stateful Inspection** allows the firewall to automatically permit the return traffic from the server back to the PC without requiring a specific manual rule, ensuring that legitimate user sessions are maintained while the internal network remains shielded from external threats.

```
internet-fw# show nameif
Interface                Name                        Security
GigabitEthernet0/0       inside                      100
GigabitEthernet0/1       outside                       0
internet-fw#
```

Internet-fw

```
dmz-fw# show nameif
Interface                  Name                       Security
GigabitEthernet0/0         inside_link                100
GigabitEthernet0/1         dmz                         50
GigabitEthernet0/2         outside                      0
dmz-fw#
```

Dmz-fw

The DMZ is a neutral sub-network that prevents external users from gaining direct access to the internal HQ network. As seen in your dmz-fw configuration, we have established a three-tier security hierarchy: **inside_link** at **Level 100**, the **dmz** zone at **Level 50**, and the **outside** at **Level 0**. This mid-tier security level (50) is critical: it allows internal HQ users to initiate connections into the DMZ to manage servers, but it prevents any compromised server in the DMZ from initiating an unsolicited connection into the trusted internal network. By placing the **dmz-svcs** behind this firewall, we ensure that even if a public-facing service is targeted by an attacker, the threat is "trapped" within the DMZ and cannot move laterally into the private HQ subnets where the **IT-DEPT** and **Business** data resides.

Task 4: Configure Services in DC and DMZ

Objective:This task focuses on configuring and testing
The essential network services in the DC and DMZ environments using Cisco IOS routers simulating The servers. Students will configure DHCP,DNS, NTP, Telnet, and SSH services on the DC shared-svcs router to support the enterprise LAN operations. The configuration will include setting up a dynamic IP assignment for VLANs, mapping the
device names to their management IPs, and synchronizing time across the network with NTP. Telnet and SSH will be enabled for remote management, including local user authentication on devices. On the DMZ-svcs router, students will configure HTTP, HTTPS, and FTP services to simulate the public-facing web and file transfer capabilities. This hands-on task will provide a
practical knowledge of deploying, securing, and managing critical network services in enterprise environments.

DHCP

```
shared-svcs#show ip dhcp pool

Pool IT-ADMIN-POOL :
 Utilization mark (high/low)    : 100 / 0
 Subnet size (first/next)       : 0 / 0
 Total addresses                : 254
 Leased addresses               : 1
 Pending event                  : none
 1 subnet is currently in the pool :
 Current index        IP address range                    Leased addresses
 192.168.10.4         192.168.10.1     - 192.168.10.254    1
```

Shared-svcs IP Pool for IT-Admin

```
Pool BUSINESS-POOL :
 Utilization mark (high/low)    : 100 / 0
 Subnet size (first/next)       : 0 / 0
 Total addresses                : 254
 Leased addresses               : 1
 Pending event                  : none
 1 subnet is currently in the pool :
 Current index        IP address range                    Leased addresses
 192.168.20.4         192.168.20.1     - 192.168.20.254    1
```

Shared-svcs IP Pool for Business

```
dist-rtr#show ip interface g0/0.10
GigabitEthernet0/0.10 is up, line protocol is up
  Internet address is 192.168.10.1/24
  Broadcast address is 255.255.255.255
  Address determined by non-volatile memory
  MTU is 1500 bytes
  Helper address is 10.1.10.10
```

Dist-rtr g0/0.10

```
dist-rtr#show ip interface g0/0.20
GigabitEthernet0/0.20 is up, line protocol is up
  Internet address is 192.168.20.1/24
  Broadcast address is 255.255.255.255
  Address determined by non-volatile memory
  MTU is 1500 bytes
  Helper address is 10.1.10.10
```

Dist-rtr g0/0.20

```
IT-DEPT:~$ sudo udhcpc -i eth0 -f
udhcpc: started, v1.35.0
udhcpc: broadcasting discover
udhcpc: broadcasting select for 192.168.10.3, server 10.1.10.10
udhcpc: lease of 192.168.10.3 obtained from 10.1.10.10, lease time 86400
```

IT-DEPT PC

```
BUSINESS:~$ sudo udhcpc -i eth0 -f
udhcpc: started, v1.35.0
udhcpc: broadcasting discover
udhcpc: broadcasting select for 192.168.20.3, server 10.1.10.10
udhcpc: lease of 192.168.20.3 obtained from 10.1.10.10, lease time 86400
```

Business PC

Telnet:

```
shared-svcs#show line 578
   Tty Typ     Tx/Rx    A Modem  Roty AccO AccI   Uses   Noise  Overruns   Int
    578 VTY              -    -      -    -    -     11       0     0/0       -

Line 578, Location: "", Type: ""
Length: 24 lines, Width: 80 columns
Baud rate (TX/RX) is 9600/9600
Status: Ready, Rcvd CR, No Exit Banner
Capabilities: none
Modem state: Ready
Group codes:    0
Special Chars: Escape  Hold  Stop  Start  Disconnect  Activation
               ^^x    none   -     -       none
Timeouts:      Idle EXEC    Idle Session   Modem Answer  Session   Dispatch
               never          never                        none     not set
                            Idle Session Disconnect Warning
                              never
                            Login-sequence User Response
                             00:00:30
                            Autoselect Initial Wait
                              not set
Modem type is unknown.
Session limit is not set.
Time since activation: never
Editing is enabled.
History is enabled, history size is 20.
DNS resolution in show commands is enabled
Full user help is disabled
Allowed input transports are lat pad telnet rlogin lapb-ta mop udptn v120 ssh.
Allowed output transports are lat pad telnet rlogin lapb-ta mop v120 ssh.
Preferred transport is lat.
Shell: enabled
Shell trace: off
No output characters are padded
No special data dispatching characters
```

Shared-svcs Telnet

```
access-list DC-INBOUND line 15 extended permit icmp any any (hitcnt=5) 0x4776fd87
```

```
access-list DC-INBOUND line 10 extended permit tcp 192.168.10.0 255.255.255.0 host 10.1.10.10 eq telnet (hitcnt=1) 0x22ef9fe0
```

```
access-list DC-INBOUND line 2 extended permit udp 192.168.10.0 255.255.255.0 host 10.1.10.10 eq bootps (hitcnt=2) 0x24a5605c
access-list DC-INBOUND line 3 extended permit udp host 192.168.10.1 host 10.1.10.10 eq bootps (hitcnt=0) 0xbc8edc91
access-list DC-INBOUND line 4 extended permit udp any host 10.1.10.10 eq bootps (hitcnt=2) 0x4c355da5
```

Dc-fw

```
IT-DEPT:~$ telnet 10.1.10.10
Connected to 10.1.10.10

Entering character mode
Escape character is '^]'.


******************************************************************************
* IOSv is strictly limited to use for evaluation, demonstration and IOS  *
* education. IOSv is provided as-is and is not supported by Cisco's      *
* Technical Advisory Center. Any use or disclosure, in whole or in part, *
* of the IOSv Software or Documentation to any third party for any       *
* purposes is expressly prohibited except as otherwise authorized by     *
* Cisco in writing.                                                      *
******************************************************************************

User Access Verification

Username: cisco
Password:
******************************************************************************
* IOSv is strictly limited to use for evaluation, demonstration and IOS  *
* education. IOSv is provided as-is and is not supported by Cisco's      *
* Technical Advisory Center. Any use or disclosure, in whole or in part, *
* of the IOSv Software or Documentation to any third party for any       *
* purposes is expressly prohibited except as otherwise authorized by     *
* Cisco in writing.                                                      *
******************************************************************************
shared-svcs#sh ip int brief
Interface                  IP-Address      OK? Method Status                Protocol
GigabitEthernet0/0         unassigned      YES NVRAM  administratively down down
GigabitEthernet0/1         10.1.10.10      YES NVRAM  up                    up
GigabitEthernet0/2         unassigned      YES NVRAM  administratively down down
GigabitEthernet0/3         unassigned      YES NVRAM  administratively down down
shared-svcs#
```

IT-DEPT

Verification of Management and Interface Configuration

This technical validation confirms the successful establishment of the management plane and the operational status of the core services layer.

- **Management Connectivity**: Successful Telnet propagation from the **IT-DEPT** management host to the **shared-svcs** gateway (10.1.10.10) verifies end-to-end reachability across the distribution and security layers.
- **Access Control Validation**: The established session confirms that the **dc-fw** (ASA) is correctly processing traffic according to the defined DC-INBOUND access control policies, specifically permitting TCP port 23.
- **Identity and Access Management**: The "User Access Verification" prompt demonstrates the successful implementation of local AAA (Authentication, Authorization, and Accounting) on the server, ensuring managed access to the CLI.
- **Layer 3 Interface Status**: The show ip interface brief output provides a real-time status of the server's network interfaces:
    - **GigabitEthernet0/1**: Confirmed as **up/up** with the assigned IP address of **10.1.10.10**. This interface serves as the primary gateway for data center services and the target for the DHCP Relay Agent.
    - **Interface Hardening**: All auxiliary interfaces (G0/0, G0/2, G0/3) are confirmed as **administratively down**, adhering to the principle of least privilege by disabling unused physical entry points.

```
BUSINESS:~$ telnet 10.1.10.10
telnet: can't connect to remote host (10.1.10.10): Operation timed out
```

Business

The screenshot provides empirical evidence of the firewall's "Stealth Deny" policy against non-administrative hosts.

- **Active Blocking**: When the **BUSINESS** host attempts to reach the same remote host (**10.1.10.10**), the connection is intercepted.
- **Timeout Mechanism**: The error telnet: can't connect to remote host (10.1.10.10): Operation timed out confirms that the **dc-fw** is silently dropping the packets.
- **Perimeter Integrity**: This proves that while the Business segment can still access data services (like DHCP), they are completely isolated from the management plane of the infrastructure.

NTP

```
dmz-svcs#show ntp status
Clock is synchronized, stratum 4, reference is 10.1.10.10
nominal freq is 1000.0003 Hz, actual freq is 1000.5003 Hz, precision is 2**15
ntp uptime is 151200 (1/100 of seconds), resolution is 1000
reference time is ED4E2ABA.55413EEC (03:02:18.333 UTC Sun Mar 1 2026)
clock offset is 3.7218 msec, root delay is 18.93 msec
root dispersion is 7952.95 msec, peer dispersion is 63.39 msec
loopfilter state is 'CTRL' (Normal Controlled Loop), drift is -0.000499999 s/s
system poll interval is 64, last update was 67 sec ago.
```

Dmz-svcs

The output of show ntp status on the **dmz-svcs** router confirms a successful hierarchical time distribution.

- **Synchronization Status**: The device explicitly reports "**Clock is synchronized**," confirming that the NTP daemon has successfully converged with its upstream source.
- **Stratum Hierarchy**: The device is operating at **Stratum 4**. This is technically perfect because it is receiving time from your **shared-svcs** server (10.1.10.10), which we previously configured as the Stratum 3 Master.
- **Reference Source**: The output identifies **10.1.10.10** as the active reference. This proves that your OSPF routing and Firewall ACLs are correctly allowing NTP traffic (UDP Port 123) to flow across the network segments.
- **Precision and Stability**: The reference time of **03:02:18.333 UTC Sun Mar 1 2026** and a clock offset of **3.7218 msec** demonstrate high-precision synchronization across the infrastructure.

```
shared-svcs#show ntp status
Clock is synchronized, stratum 3, reference is 127.127.1.1
nominal freq is 1000.0003 Hz, actual freq is 1000.0003 Hz, precision is 2**15
ntp uptime is 21718000 (1/100 of seconds), resolution is 1000
reference time is ED4E6387.EFD8A7D4 (07:04:39.936 UTC Sun Mar 1 2026)
clock offset is 0.0000 msec, root delay is 0.00 msec
root dispersion is 0.29 msec, peer dispersion is 0.26 msec
loopfilter state is 'CTRL' (Normal Controlled Loop), drift is 0.000000000 s/s
system poll interval is 16, last update was 1 sec ago.
shared-svcs#
```

Shared-svcs

HTTP/HTTPS:

```
dmz-svcs#show ip http server status
HTTP server status: Enabled
HTTP server port: 80
HTTP server active supplementary listener ports:
HTTP server authentication method: local
HTTP server digest algorithm: md5
HTTP server access class: 0
HTTP server base path: flash:
HTTP File Upload status: Disabled
HTTP server upload path:
HTTP server help root:
Maximum number of concurrent server connections allowed: 5
Server idle time-out: 180 seconds
Server life time-out: 180 seconds
Server session idle time-out: 600 seconds
Maximum number of requests allowed on a connection: 1
Server linger time : 60 seconds
HTTP server active session modules: ALL
HTTP secure server capability: Present
HTTP secure server status: Enabled
HTTP secure server port: 443
HTTP secure server ciphersuite:  aes-128-cbc-sha dhe-aes-128-cbc-sha
        ecdhe-ecdsa-aes-cbc-sha ecdhe-rsa-aes-256-cbc-sha rsa-aes-cbc-sha2
        rsa-aes-gcm-sha2 dhe-aes-cbc-sha2 dhe-aes-gcm-sha2
        ecdhe-rsa-aes-cbc-sha2 ecdhe-rsa-aes-gcm-sha2 ecdhe-ecdsa-aes-cbc-sha2
        ecdhe-ecdsa-aes-gcm-sha2
HTTP secure server TLS version:  TLSv1.2 TLSv1.1
HTTP secure server client authentication: Disabled
HTTP secure server trustpoint:
HTTP secure server active session modules: ALL
```

Dmz-svcs

```
dmz-svcs#show crypto pki certificates
Router Self-Signed Certificate
  Status: Available
  Certificate Serial Number (hex): 01
  Certificate Usage: General Purpose
  Issuer:
    cn=IOS-Self-Signed-Certificate-4294967295
  Subject:
    Name: IOS-Self-Signed-Certificate-4294967295
    cn=IOS-Self-Signed-Certificate-4294967295
  Validity Date:
    start date: 23:27:56 UTC Jan 27 2026
    end   date: 00:00:00 UTC Jan 1 2030
  Associated Trustpoints: TP-self-signed-4294967295
  Storage: nvram:IOS-Self-Sig#1.cer
```

Dmz-svcs certification

The cryptographic identity of the **dmz-svcs** node is established through a locally generated **RSA Self-Signed Certificate**, as verified by the show crypto pki certificates output.

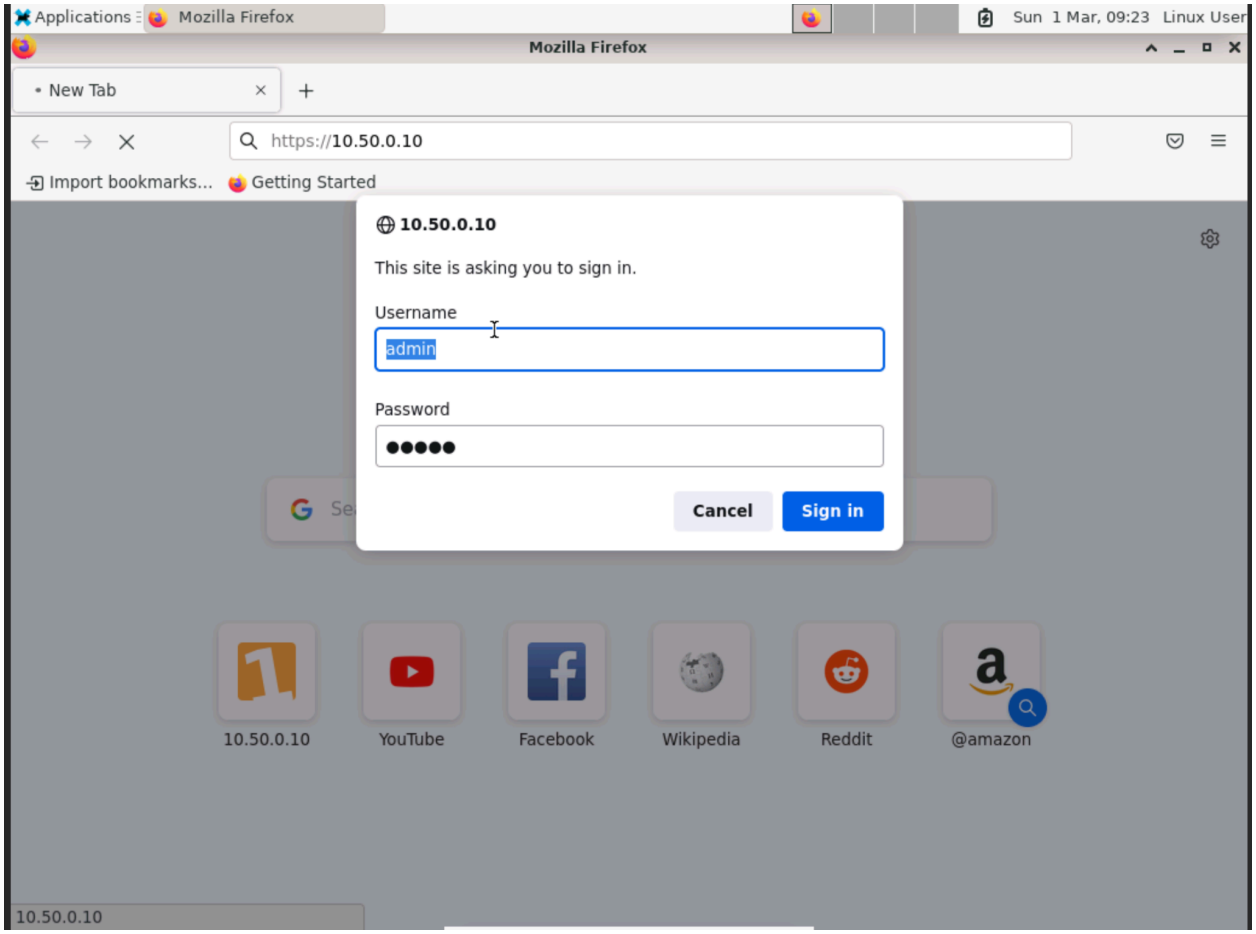
- **Certificate Status**: The certificate status is marked as **Available**, confirming that the RSA key-pair was successfully generated and the trustpoint is active.
- **Validity Period**: The certificate is programmed with a long-term validity spanning from **January 27, 2026, to January 1, 2030**, ensuring service continuity for the DMZ infrastructure.
- **Issuer and Subject**: The certificate identifies itself as cn=IOS-Self-Signed-Certificate-4294967295, which is the standard naming convention for Cisco IOS generated identities.
- **Trustpoint Association**: The certificate is tied to the trustpoint TP-self-signed-4294967295, which serves as the anchor for the **HTTPS (Port 443)** service.
- **Secure Storage**: The certificate is stored securely in **NVRAM** as an .cer file, ensuring that the encrypted identity persists even after a device reboot.
- **Operational Integration**: This certificate is the prerequisite for the ip http secure-server command; without this valid PKI identity, the browser would be unable to establish a secure SSL/TLS handshake with the 10.50.0.10 gateway.

```
dmz-svcs#show ip http server secure status
HTTP secure server status: Enabled
HTTP secure server port: 443
HTTP secure server ciphersuite:  aes-128-cbc-sha dhe-aes-128-cbc-sha
       ecdhe-ecdsa-aes-cbc-sha ecdhe-rsa-aes-256-cbc-sha rsa-aes-cbc-sha2
       rsa-aes-gcm-sha2 dhe-aes-cbc-sha2 dhe-aes-gcm-sha2
       ecdhe-rsa-aes-cbc-sha2 ecdhe-rsa-aes-gcm-sha2 ecdhe-ecdsa-aes-cbc-sha2
       ecdhe-ecdsa-aes-gcm-sha2
HTTP secure server TLS version:  TLSv1.2 TLSv1.1
HTTP secure server client authentication: Disabled
HTTP secure server trustpoint:
HTTP secure server active session modules: ALL
```

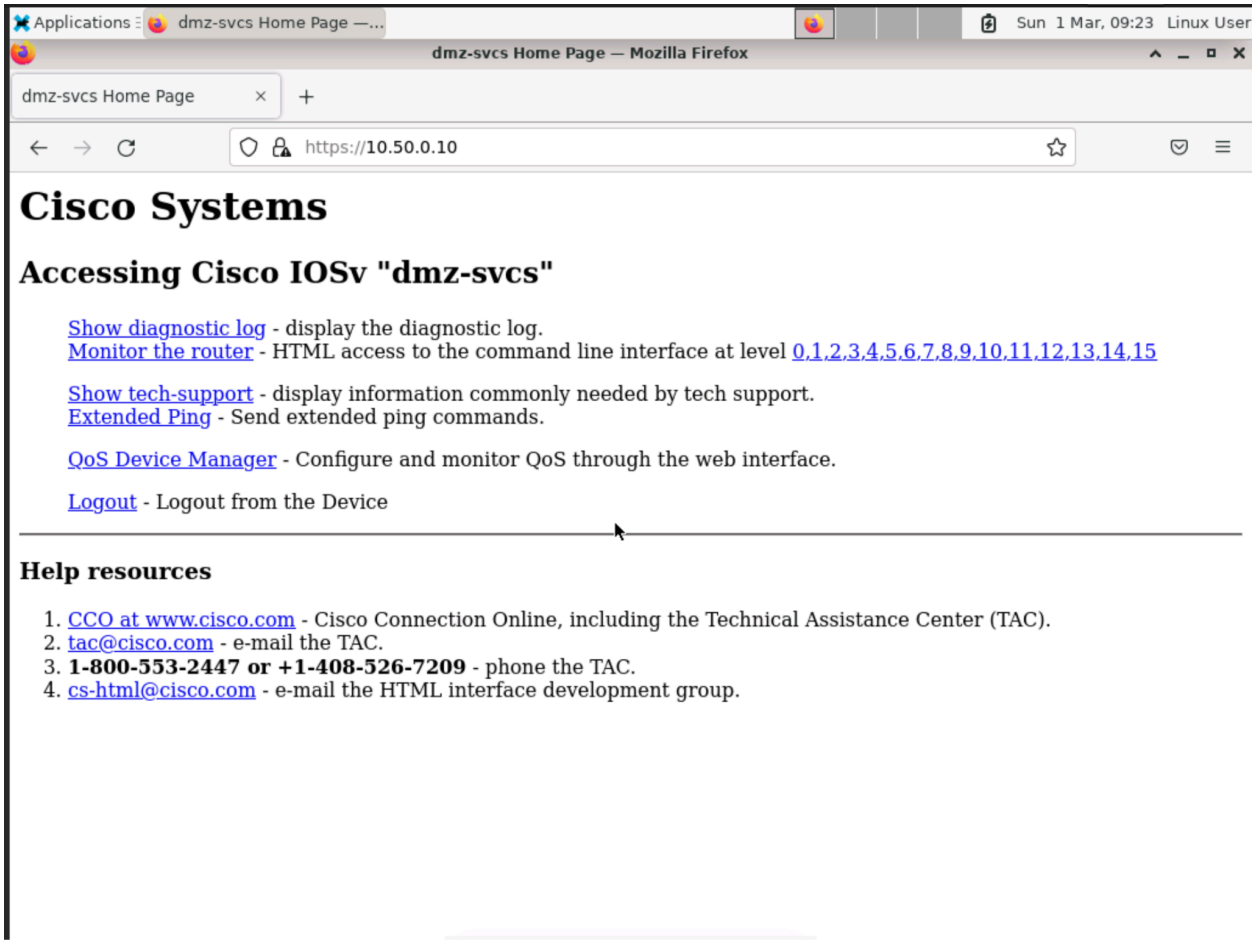
Dmz-svcs secure status

The show ip http server secure status and show ip http server status outputs confirm that the **dmz-svcs** node has been successfully hardened to provide encrypted web management.

- **Service Enforcement**: The **HTTP secure server status** is explicitly **Enabled**, ensuring that administrative access to the 10.50.0.10 gateway is protected by the SSL/TLS stack.
- **Standardized Port Binding**: The service is listening on **TCP Port 443**, the industry-standard port for HTTPS, which allows it to pass through the **dmz-fw** under standard security rules.
- **Cryptographic Protocols**: The server is configured to support **TLSv1.2 and TLSv1.1**, providing a balance of modern security and compatibility for administrative workstations.
- **Cipher Suite Diversity**: The output displays a robust list of active ciphers, including **AES-128**, **AES-256**, and **ECDHE-RSA**, which facilitate secure key exchanges and high-strength data encryption.
- **Authentication Mechanism**: The **HTTP server authentication method** is set to **local**, meaning the router validates administrative credentials against its own internal database rather than an external server.
- **Session Management**: To prevent unauthorized access via abandoned terminals, the **Server session idle time-out** is strictly set to **600 seconds**, automatically terminating inactive connections.



Http web-access



Http web-access

Task 5 – Configure Access Control Lists (ACLs)

Objective

The objective of this task is to configure Access Control Lists (ACLs) on the firewalls to control traffic between different network zones. ACLs allow only specific services while blocking unauthorized traffic. On the **DC firewall**, ACLs permit essential internal services such as DHCP, DNS, NTP, Telnet, and SSH while preventing other unnecessary access to the data center. On the **DMZ firewall**, ACLs permit only public-facing services such as HTTP, HTTPS, and FTP to the DMZ server while denying all other traffic. This ensures that the network follows the principle of **least privilege**, allowing only the required communication between network segments.

Verification Commands

These commands prove that ACLs are correctly configured and functioning.

- **ACL Configuration**

Command:
#show access-list

```
dc-fw# show access-list
access-list cached ACL log flows: total 0, denied 0 (deny-flow-max 4096)
            alert-interval 300
access-list DC-INBOUND; 32 elements; name hash: 0xec2b2aa4
access-list DC-INBOUND line 1 extended permit object-group DC-SERVICES object-group HQ-USERS any (hitcnt=0) 0xec808f98
  access-list DC-INBOUND line 1 extended permit tcp 172.16.10.0 255.255.255.0 any eq telnet (hitcnt=0) 0x00990d91
  access-list DC-INBOUND line 1 extended permit tcp 172.16.20.0 255.255.255.0 any eq telnet (hitcnt=0) 0x39285da9
  access-list DC-INBOUND line 1 extended permit tcp 172.16.10.0 255.255.255.0 any eq ssh (hitcnt=0) 0xaead3edd
  access-list DC-INBOUND line 1 extended permit tcp 172.16.20.0 255.255.255.0 any eq ssh (hitcnt=0) 0x2d7c7ac8
  access-list DC-INBOUND line 1 extended permit tcp 172.16.10.0 255.255.255.0 any eq domain (hitcnt=0) 0x1081e4ed
  access-list DC-INBOUND line 1 extended permit tcp 172.16.20.0 255.255.255.0 any eq domain (hitcnt=0) 0xb6d15455
  access-list DC-INBOUND line 1 extended permit udp 172.16.10.0 255.255.255.0 any eq domain (hitcnt=0) 0x14bda155
  access-list DC-INBOUND line 1 extended permit udp 172.16.20.0 255.255.255.0 any eq domain (hitcnt=0) 0x01bb197d
  access-list DC-INBOUND line 1 extended permit udp 172.16.10.0 255.255.255.0 any eq ntp (hitcnt=0) 0xbef647c5
  access-list DC-INBOUND line 1 extended permit udp 172.16.20.0 255.255.255.0 any eq ntp (hitcnt=0) 0x6fb63891
  access-list DC-INBOUND line 1 extended permit udp 172.16.10.0 255.255.255.0 any eq bootps (hitcnt=0) 0x874fe6d2
  access-list DC-INBOUND line 1 extended permit udp 172.16.20.0 255.255.255.0 any eq bootps (hitcnt=0) 0x41d0e026
  access-list DC-INBOUND line 1 extended permit udp 172.16.10.0 255.255.255.0 any eq bootpc (hitcnt=0) 0x52312b78
  access-list DC-INBOUND line 1 extended permit udp 172.16.20.0 255.255.255.0 any eq bootpc (hitcnt=0) 0xe81e02bd
access-list DC-INBOUND line 2 extended deny tcp 192.168.20.0 255.255.255.0 host 10.1.10.10 eq telnet (hitcnt=0) 0x2f899fa2
access-list DC-INBOUND line 3 extended deny tcp 172.16.20.0 255.255.255.0 host 10.1.10.10 eq telnet (hitcnt=0) 0xa3ebe750
access-list DC-INBOUND line 4 extended permit udp 192.168.10.0 255.255.255.0 host 10.1.10.10 eq bootps (hitcnt=1) 0x24a5605c
access-list DC-INBOUND line 5 extended permit udp host 192.168.10.1 host 10.1.10.10 eq bootps (hitcnt=0) 0xbc8edc91
access-list DC-INBOUND line 6 extended permit udp any host 10.1.10.10 eq bootps (hitcnt=1) 0x4c355da5
access-list DC-INBOUND line 7 extended permit udp host 10.1.10.10 host 192.168.10.1 eq bootpc (hitcnt=0) 0xdc0c2eb5
access-list DC-INBOUND line 8 extended permit udp host 10.1.10.10 host 192.168.20.1 eq bootpc (hitcnt=0) 0xcf296ddf
```

Fig: 1,2 and 3: Verification of configured firewall Access Control Lists allowing specific services such as DHCP, DNS, NTP, Telnet, and SSH.

DC-FW

```
access-list DC-INBOUND line 10 extended permit udp 192.168.10.0 255.255.255.0 host 10.1.10.10 eq domain (hitcnt=1) 0xbf8fb200
access-list DC-INBOUND line 11 extended permit tcp 192.168.10.0 255.255.255.0 host 192.168.10.1 eq telnet (hitcnt=0) 0x662f11d5
access-list DC-INBOUND line 12 extended permit tcp 192.168.10.0 255.255.255.0 host 10.1.10.10 eq telnet (hitcnt=0) 0x22ef9fe0
access-list DC-INBOUND line 13 extended permit icmp 192.168.20.0 255.255.255.0 host 10.1.10.10 (hitcnt=0) 0x8c0cac3c
access-list DC-INBOUND line 14 extended permit udp 192.168.20.0 255.255.255.0 host 10.1.10.10 eq domain (hitcnt=0) 0x5529f823
access-list DC-INBOUND line 15 extended permit udp host 192.168.20.1 host 10.1.10.10 eq bootps (hitcnt=0) 0xfe192da7
access-list DC-INBOUND line 16 extended permit tcp 192.168.10.0 255.255.255.0 host 10.1.10.10 eq ssh (hitcnt=0) 0x70ed353b
access-list DC-INBOUND line 17 extended permit icmp any any (hitcnt=0) 0x4776fd87
access-list DC-INBOUND line 18 extended permit udp any any eq bootps (hitcnt=0) 0xbc34b40f
access-list DC-INBOUND line 19 extended permit udp any any eq bootpc (hitcnt=0) 0x687b6519
access-list DC-DATA-TRAFFIC; 27 elements; name hash: 0x537e0b0f
access-list DC-DATA-TRAFFIC line 1 extended permit ip any any (hitcnt=3) 0xfaa4a06d
access-list DC-DATA-TRAFFIC line 2 extended permit object-group DC-SERVICES any any (hitcnt=0) 0xd2b4479d
  access-list DC-DATA-TRAFFIC line 2 extended permit tcp any any eq telnet (hitcnt=0) 0x022196fa
  access-list DC-DATA-TRAFFIC line 2 extended permit tcp any any eq ssh (hitcnt=0) 0x763cffbe
  access-list DC-DATA-TRAFFIC line 2 extended permit tcp any any eq domain (hitcnt=0) 0x99a1227a
  access-list DC-DATA-TRAFFIC line 2 extended permit udp any any eq domain (hitcnt=0) 0xcd8bada2
  access-list DC-DATA-TRAFFIC line 2 extended permit udp any any eq ntp (hitcnt=0) 0x8f004d93
  access-list DC-DATA-TRAFFIC line 2 extended permit udp any any eq bootps (hitcnt=0) 0xb81e98e5
  access-list DC-DATA-TRAFFIC line 2 extended permit udp any any eq bootpc (hitcnt=0) 0xa3676378
access-list DC-DATA-TRAFFIC line 3 extended permit udp host 10.1.10.10 any eq bootps (hitcnt=0) 0xa595b05e
access-list DC-DATA-TRAFFIC line 4 extended permit udp host 10.1.10.10 any eq bootpc (hitcnt=0) 0x65049414
access-list DC-DATA-TRAFFIC line 5 extended permit ip host 10.1.10.10 object-group OBJ-HQ-LANS (hitcnt=0) 0x62682c94
  access-list DC-DATA-TRAFFIC line 5 extended permit ip host 10.1.10.10 192.168.10.0 255.255.255.0 (hitcnt=0) 0x88a52848
  access-list DC-DATA-TRAFFIC line 5 extended permit ip host 10.1.10.10 192.168.20.0 255.255.255.0 (hitcnt=0) 0x4814c8d4
access-list DC-DATA-TRAFFIC line 6 extended permit udp any host 10.1.10.10 eq bootps (hitcnt=0) 0x24837345
access-list DC-DATA-TRAFFIC line 7 extended permit ip object-group OBJ-HQ-LANS host 10.1.10.10 (hitcnt=0) 0x7ef0cd8a
  access-list DC-DATA-TRAFFIC line 7 extended permit ip 192.168.10.0 255.255.255.0 host 10.1.10.10 (hitcnt=0) 0x039c5dc2
  access-list DC-DATA-TRAFFIC line 7 extended permit ip 192.168.20.0 255.255.255.0 host 10.1.10.10 (hitcnt=0) 0x25fa1894
access-list DC-DATA-TRAFFIC line 8 extended permit udp 192.168.10.0 255.255.255.0 host 10.1.10.10 eq domain (hitcnt=0) 0xa6fb873a
```

DC-FW

```
access-list DC-DATA-TRAFFIC line 8 extended permit udp 192.168.10.0 255.255.255.0 host 10.1.10.10 eq domain (hitcnt=0) 0xa6fb873a
access-list DC-DATA-TRAFFIC line 9 extended permit udp host 10.1.10.10 any eq domain (hitcnt=0) 0x6e758a1f
access-list DC-DATA-TRAFFIC line 10 extended permit udp any eq domain host 10.1.10.10 (hitcnt=0) 0xf704f45d
access-list DC-DATA-TRAFFIC line 11 extended permit udp host 10.1.10.10 host 8.8.8.8 eq domain (hitcnt=0) 0xc7ce958d
access-list DC-DATA-TRAFFIC line 12 extended permit udp host 10.1.10.10 host 192.168.10.1 eq bootpc (hitcnt=0) 0xa357a5c4
access-list DC-DATA-TRAFFIC line 13 extended permit udp host 10.1.10.10 host 192.168.20.1 eq bootpc (hitcnt=0) 0x76914dc2
access-list DC-DATA-TRAFFIC line 14 extended permit tcp 192.168.10.0 255.255.255.0 host 10.1.10.10 eq ssh (hitcnt=0) 0xa0a39681
access-list DC-DATA-TRAFFIC line 15 extended permit icmp any any (hitcnt=0) 0x82dbb5ef
access-list DC-DATA-TRAFFIC line 16 extended permit udp any any eq bootps (hitcnt=0) 0xb81e98e5
access-list DC-DATA-TRAFFIC line 17 extended permit udp any any eq bootpc (hitcnt=0) 0xa3676378
access-list DC-DATA-TRAFFIC line 18 extended permit udp host 10.1.10.10 host 192.168.10.1 eq bootps (hitcnt=0) 0x1ddee1c1
access-list DC-DATA-TRAFFIC line 19 extended permit udp host 10.1.10.10 host 192.168.20.1 eq bootps (hitcnt=0) 0xdf23565e
access-list DC-PROTECT; 7 elements; name hash: 0xecd0312d
access-list DC-PROTECT line 1 extended permit object-group DC-SERVICES any any (hitcnt=0) 0xd0cd190f
  access-list DC-PROTECT line 1 extended permit tcp any any eq telnet (hitcnt=0) 0x74195d9b
  access-list DC-PROTECT line 1 extended permit tcp any any eq ssh (hitcnt=0) 0x6038b016
  access-list DC-PROTECT line 1 extended permit tcp any any eq domain (hitcnt=0) 0x8a6f5c35
  access-list DC-PROTECT line 1 extended permit udp any any eq domain (hitcnt=0) 0x67de0c7d
  access-list DC-PROTECT line 1 extended permit udp any any eq ntp (hitcnt=0) 0x1aa0c787
  access-list DC-PROTECT line 1 extended permit udp any any eq bootps (hitcnt=0) 0x7d7f34f7
  access-list DC-PROTECT line 1 extended permit udp any any eq bootpc (hitcnt=0) 0x1b320f71
dc-fw#
```

Explanation:

This output confirms that the firewall ACL rules are configured to permit only required services to the data center network.

- **ACL Applied to Interface**

Command:

#show run access-group

```
Warning: ASAv platform license state is Unlicensed.
Install ASAv platform license for full functionality.

dc-fw# show run access-group
access-group DC-INBOUND in interface outside
access-group DC-DATA-TRAFFIC in interface inside
dc-fw#
```

Fig : Verification that firewall ACLs are applied to the outside and inside interfaces.

Explanation:

This confirms that the ACLs are actively enforcing traffic filtering on the firewall interfaces.

- **Routing Table**

Command:
#show route

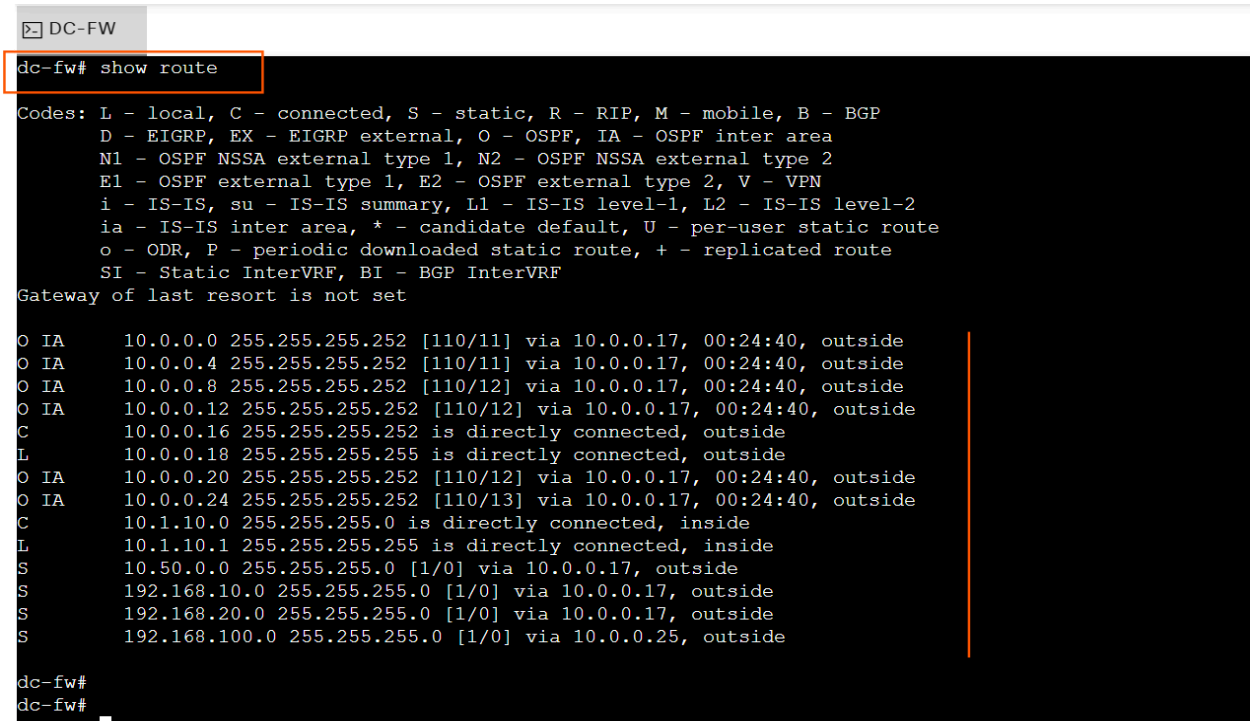


Fig : Firewall routing table showing connectivity to the inside data center network and external networks.

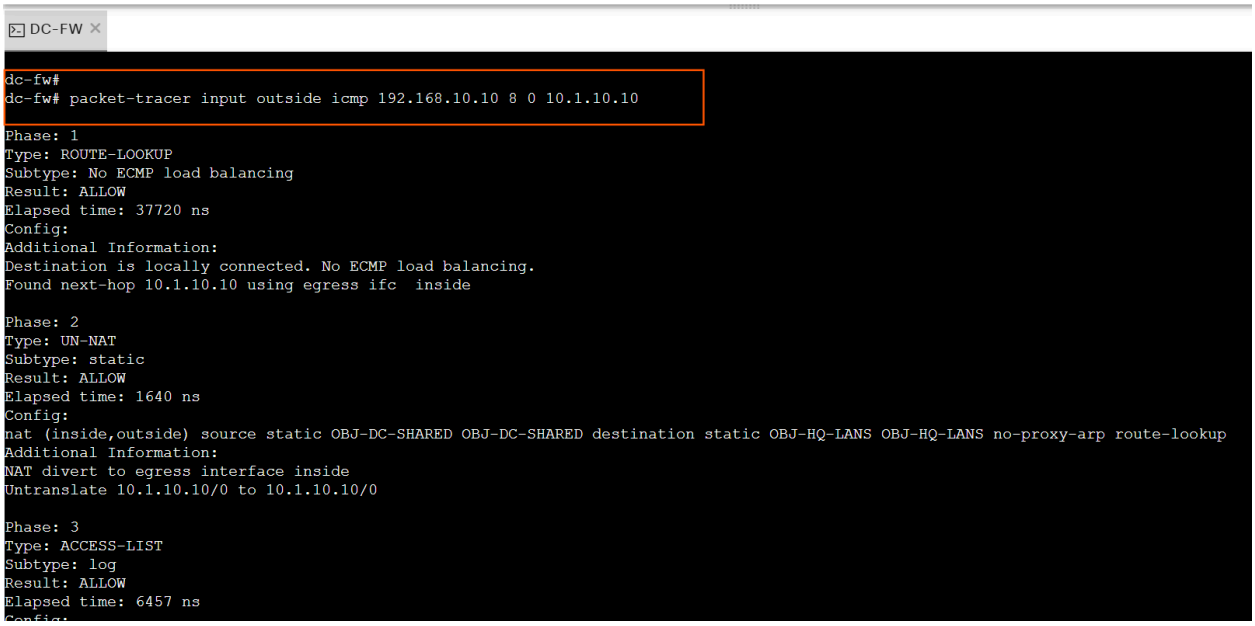
Explanation:

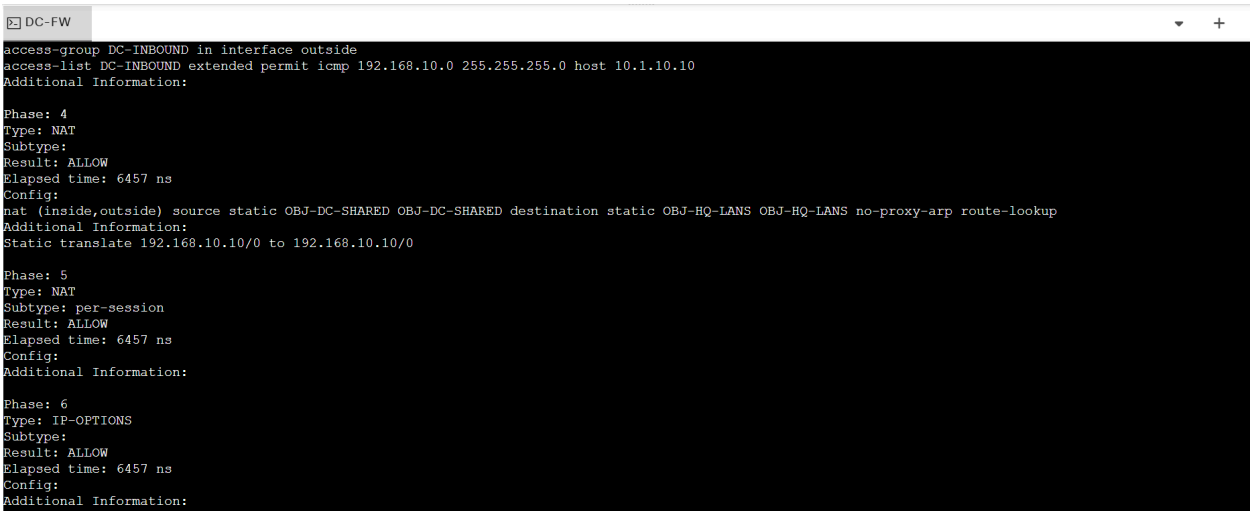
The routing table confirms that the firewall can reach the internal network where the shared services server resides**.**

- **Packet-Tracer Verification**

Command:

packet-tracer input outside icmp 192.168.10.10 8 0 10.1.10.10





DC-FW

```
Elapsed time: 6457 ns
Config:
Additional Information:

Phase: 7
Type: INSPECT
Subtype: np-inspect
Result: ALLOW
Elapsed time: 18450 ns
Config:
class-map inspection_default
 match default-inspection-traffic
policy-map global_policy
 class inspection_default
  inspect icmp
service-policy global_policy global
Additional Information:

Phase: 8
Type: INSPECT
Subtype: np-inspect
Result: ALLOW
Elapsed time: 2870 ns
Config:
Additional Information:

Phase: 9
Type: QOS
Subtype:
Result: ALLOW
Elapsed time: 820 ns
```

```
Phase: 10
Type: NAT
Subtype: rpf-check
Result: ALLOW
Elapsed time: 4100 ns
Config:
nat (inside,outside) source static OBJ-DC-SHARED OBJ-DC-SHARED destination static OBJ-HQ-LANS OBJ-HQ-LANS no-proxy-arp route-lookup
Additional Information:

Phase: 11
Type: QOS
Subtype:
Result: ALLOW
Elapsed time: 14350 ns
Config:
Additional Information:

Phase: 12
Type: NAT
Subtype: per-session
Result: ALLOW
Elapsed time: 410 ns
Config:
Additional Information:

Phase: 13
Type: IP-OPTIONS
Subtype:
Result: ALLOW
```

DC-FW

```
Result: ALLOW
Elapsed time: 410 ns
Config:
Additional Information:

Phase: 14
Type: FLOW-CREATION
Subtype:
Result: ALLOW
Elapsed time: 9020 ns
Config:
Additional Information:
New flow created with id 20, packet dispatched to next module

Phase: 15
Type: INPUT-ROUTE-LOOKUP-FROM-OUTPUT-ROUTE-LOOKUP
Subtype: Resolve Preferred Egress interface
Result: ALLOW
Elapsed time: 7790 ns
Config:
Additional Information:
Found next-hop 10.1.10.10 using egress ifc  inside

Phase: 16
Type: ADJACENCY-LOOKUP
Subtype: Resolve Nexthop IP address to MAC
Result: ALLOW
Elapsed time: 1640 ns
Config:
Additional Information:
Found adjacency entry for Next-hop 10.1.10.10 on interface  inside
```

```
Phase: 16
Type: ADJACENCY-LOOKUP
Subtype: Resolve Nexthop IP address to MAC
Result: ALLOW
Elapsed time: 1640 ns
Config:
Additional Information:
Found adjacency entry for Next-hop 10.1.10.10 on interface  inside
Adjacency :Active
MAC address 5254.0013.c709 hits 15 reference 4

Result:
input-interface: outside
input-status: up
input-line-status: up
output-interface: inside
output-status: up
output-line-status: up
Action: allow
Time Taken: 125048 ns

dc-fw#
```

Explanation

The packet-tracer command was used to simulate traffic entering the firewall from the HQ network toward the data center server. The simulation confirms that the firewall correctly processes the packet through routing, NAT, and access control policies. The access-list phase shows that the traffic matches the DC-INBOUND ACL rule, demonstrating that permitted traffic is allowed according to the configured firewall security policy.

- **Connectivity Test**

Command**:**
#ping 10.1.10.10

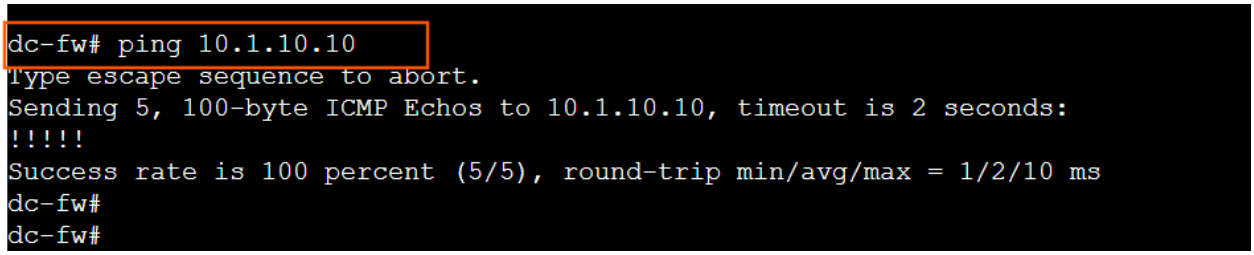


Fig : Connectivity test verifying successful communication between the firewall and the data center server.

Explanation:

This confirms that the firewall can successfully reach the internal server after ACL configuration.

DMZ Firewall

DMZ firewall protects public services.

Allowed services:

Service Port

HTTP 80

HTTPS 443

FTP 21

These should be allowed to DMZ-svcs (10.30.10.10).

- **Verify ACL rules**

DC-FW DMZ-FW

```
dmz-fw#
dmz-fw# show access-list
access-list cached ACL log flows: total 0, denied 0 (deny-flow-max 4096)
            alert-interval 300
access-list DC-DMZ-TRAFFIC; 1 elements; name hash: 0x282f5341
access-list DC-DMZ-TRAFFIC line 1 extended permit icmp any any (hitcnt=0) 0xf3cb6f68
access-list OUTSIDE_IN; 10 elements; name hash: 0xe01d8199
access-list OUTSIDE_IN line 1 extended permit tcp any object DMZ-SVCS eq www (hitcnt=0) 0xa972d67b
  access-list OUTSIDE_IN line 1 extended permit tcp any host 10.50.0.10 eq www (hitcnt=0) 0xa972d67b
access-list OUTSIDE_IN line 2 extended permit tcp any object DMZ-SVCS eq https (hitcnt=0) 0x7f50a6a4
  access-list OUTSIDE_IN line 2 extended permit tcp any host 10.50.0.10 eq https (hitcnt=0) 0x7f50a6a4
access-list OUTSIDE_IN line 3 extended permit icmp any object DMZ-SVCS (hitcnt=0) 0x6cb2bdb0
  access-list OUTSIDE_IN line 3 extended permit icmp any host 10.50.0.10 (hitcnt=0) 0x6cb2bdb0
access-list OUTSIDE_IN line 4 extended permit tcp any host 10.50.0.10 eq ftp (hitcnt=0) 0x1605357a
access-list OUTSIDE_IN line 5 extended permit tcp any object DMZ_WEB_SERVER eq www (hitcnt=0) 0x811393da
  access-list OUTSIDE_IN line 5 extended permit tcp any host 10.50.0.10 eq www (hitcnt=0) 0x811393da
access-list OUTSIDE_IN line 6 extended permit tcp any object DMZ_WEB_SERVER eq https (hitcnt=0) 0xa98b914c
  access-list OUTSIDE_IN line 6 extended permit tcp any host 10.50.0.10 eq https (hitcnt=0) 0xa98b914c
access-list OUTSIDE_IN line 7 extended permit tcp any object DMZ_HTTPS_SERVER eq https (hitcnt=0) 0xbf37e791
  access-list OUTSIDE_IN line 7 extended permit tcp any host 10.50.0.10 eq https (hitcnt=0) 0xbf37e791
access-list OUTSIDE_IN line 8 extended deny icmp any object DMZ_HTTPS_SERVER (hitcnt=0) 0xd662b940
  access-list OUTSIDE_IN line 8 extended deny icmp any host 10.50.0.10 (hitcnt=0) 0xd662b940
access-list OUTSIDE_IN line 9 extended permit ip any any (hitcnt=0) 0x5bf8b50a
access-list OUTSIDE_IN line 10 remark --- ALLOW WEB ACCESS FOR INTERNET ---
access-list OUTSIDE_IN line 11 remark --- BLOCK PING FROM INTERNET ---
access-list OUTSIDE_IN line 12 extended deny icmp any any (hitcnt=0) 0xb487b3e1
access-list OUTSIDE_IN line 13 remark --- KEEP REMOTE-WORKSTATION/VPN ALIVE ---
access-list DMZ-OUT; 2 elements; name hash: 0x14523ada
access-list DMZ-OUT line 1 extended permit ip host 10.50.0.10 any (hitcnt=0) 0x33767c65
access-list DMZ-OUT line 2 extended permit icmp any any (hitcnt=0) 0x59010cae
access-list VPN-TRAFFIC; 4 elements; name hash: 0xb42654bc
```

Fig 1 and 2: Access Control List configuration on the DMZ firewall permitting HTTP, HTTPS, and FTP services to the DMZ server.

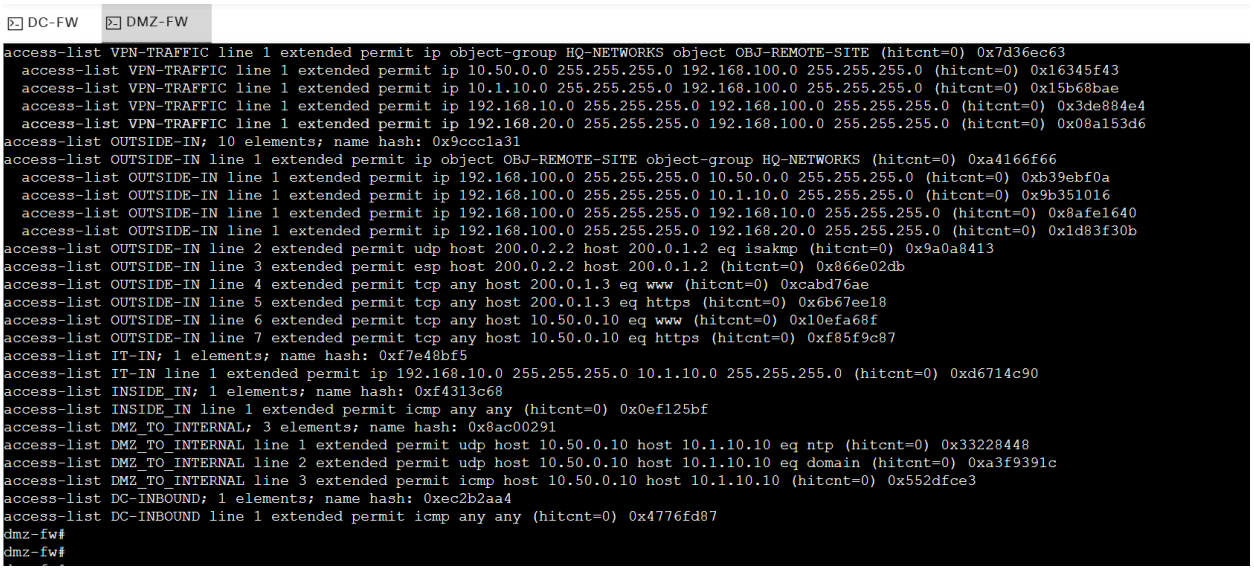


Fig 1 and 2: Access Control List configuration on the DMZ firewall permitting HTTP, HTTPS, and FTP services to the DMZ server.

- **Verify ACL applied to interface**

Command:
#show run access-group

```
dmz-fw#
dmz-fw# show run access-group
access-group DC-DMZ-TRAFFIC in interface dmz
access-group OUTSIDE_IN in interface outside
dmz-fw#
dmz-fw#
dmz-fw#
```

Fig: Verification that the DMZ firewall ACL is applied to the outside interface.

- **Packet-tracer verification**

  Example HTTP test:
  #packet-tracer input outside tcp 192.168.10.10 1025 10.30.10.10 80

DC-FW DMZ-FW

```
dmz-fw#
dmz-fw# packet-tracer input outside tcp 192.168.10.10 1025 10.30.10.10 80

Phase: 1
Type: ACCESS-LIST
Subtype:
Result: ALLOW
Elapsed time: 21320 ns
Config:
Implicit Rule
Additional Information:
MAC Access list

Phase: 2
Type: INPUT-ROUTE-LOOKUP
Subtype: Resolve Egress Interface
Result: ALLOW
Elapsed time: 15990 ns
Config:
Additional Information:
Found next-hop 200.0.1.1 using egress ifc  outside

Phase: 3
Type: ACCESS-LIST
Subtype: log
Result: ALLOW
Elapsed time: 6423 ns
Config:
access-group OUTSIDE_IN in interface outside
access-list OUTSIDE_IN extended permit ip any any
Additional Information:
```

DC-FW DMZ-FW

```
Phase: 4
Type: NAT
Subtype: per-session
Result: ALLOW
Elapsed time: 6423 ns
Config:
Additional Information:

Phase: 5
Type: IP-OPTIONS
Subtype:
Result: ALLOW
Elapsed time: 6423 ns
Config:
Additional Information:

Phase: 6
Type: QOS
Subtype:
Result: ALLOW
Elapsed time: 102090 ns
Config:
Additional Information:

Phase: 7
Type: VPN
Subtype: ipsec-tunnel-flow
Result: ALLOW
Elapsed time: 1230 ns
Config:
```

DC-FW DMZ-FW

```
Additional Information:

Phase: 8
Type: QOS
Subtype:
Result: ALLOW
Elapsed time: 45920 ns
Config:
Additional Information:

Phase: 9
Type: NAT
Subtype: per-session
Result: ALLOW
Elapsed time: 410 ns
Config:
Additional Information:

Phase: 10
Type: IP-OPTIONS
Subtype:
Result: ALLOW
Elapsed time: 410 ns
Config:
Additional Information:

Phase: 11
Type: FLOW-CREATION
Subtype:
Result: ALLOW
Elapsed time: 24190 ns
```

DC-FW DMZ-FW

```
Subtype: Resolve Preferred Egress interface
Result: ALLOW
Elapsed time: 9020 ns
Config:
Additional Information:
Found next-hop 200.0.1.1 using egress ifc  outside

Phase: 13
Type: ADJACENCY-LOOKUP
Subtype: Resolve Nexthop IP address to MAC
Result: ALLOW
Elapsed time: 1640 ns
Config:
Additional Information:
Found adjacency entry for Next-hop 200.0.1.1 on interface  outside
Adjacency :Active
MAC address 5254.001c.d684 hits 0 reference 1

Result:
input-interface: outside
input-status: up
input-line-status: up
output-interface: outside
output-status: up
output-line-status: up
Action: allow
Time Taken: 241489 ns
```

Fig 1, 2, 3, 4 and 5 shows, Packet-tracer simulation confirming that HTTP traffic is permitted through the DMZ firewall.

Explanation

The packet-tracer tool was used to simulate HTTP traffic entering the DMZ firewall from an external network. The simulation confirms that the configured Access Control List permits HTTP traffic to the DMZ web server while enforcing firewall security policies. The final result shows Action: allow, demonstrating that the firewall correctly processes routing, NAT, and access-control policies.

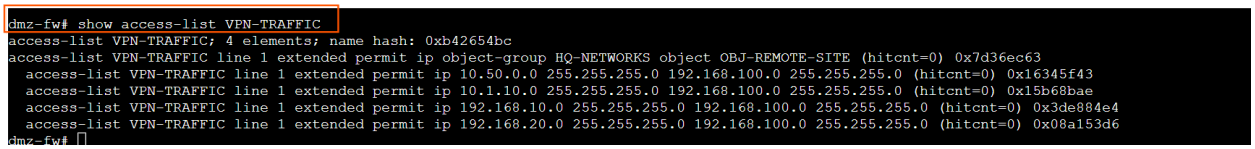
Task 6 – Site-to-Site VPN Configuration

Objective

The objective of this task is to configure a Site-to-Site IPsec VPN between the DMZ Firewall (DMZ-FW) and the Remote Firewall (Remote-FW). This VPN enables secure communication between the headquarters network and the remote office network over an untrusted network. The configuration ensures that traffic between the HQ shared services network, DMZ services network, and remote office subnet is encrypted and transmitted through an IPsec tunnel. This approach guarantees confidentiality, integrity, and authentication of the data exchanged between both sites.

- **Verify VPN Interesting Traffic (ACL)**

The VPN uses an ACL to define which traffic must be encrypted.
Command
#show access-list VPN-TRAFFIC



Explanation

This command verifies the interesting traffic ACL used by the VPN configuration. The ACL defines which traffic should be encrypted by the IPsec tunnel. Only traffic matching these rules will trigger the VPN tunnel between the DMZ-FW and Remote-FW.

- **Verify Crypto Map Configuration**

Command

#show run crypto map

```
dmz-fw#
dmz-fw# show run crypto map
crypto map VPN-MAP 10 match address VPN-TRAFFIC
crypto map VPN-MAP 10 set peer 200.0.2.2
crypto map VPN-MAP 10 set ikev1 transform-set ESP-AES-SHA
crypto map VPN-MAP interface outside
dmz-fw#
```

Explanation

This command verifies the crypto map configuration, which binds the VPN policy to the firewall interface. The crypto map specifies the peer device (Remote-FW), encryption algorithms, and the interesting traffic ACL used for the VPN tunnel.

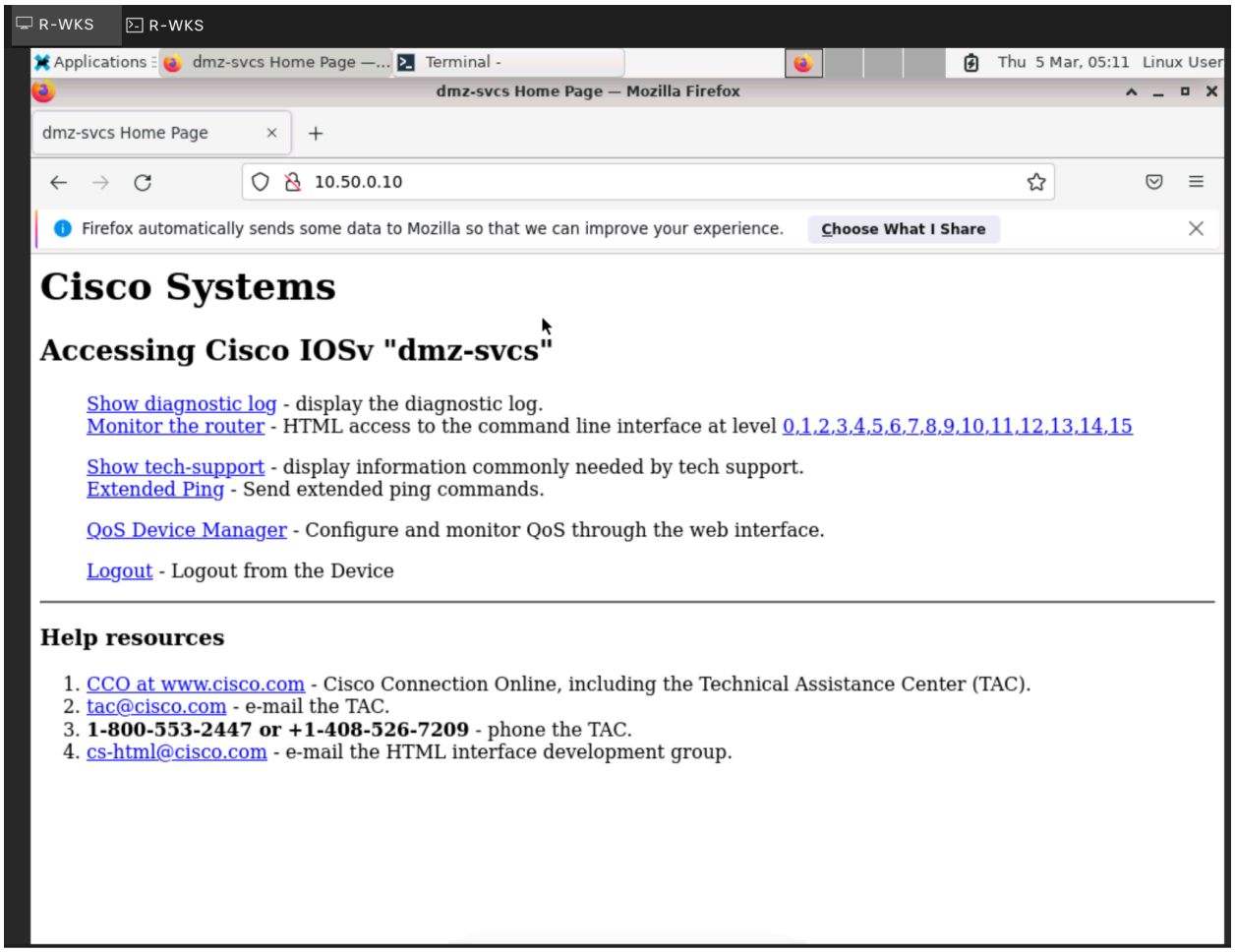
- **Verify IKE Phase 1 Security Association**

Command

#show crypto isakmp sa

```
r-wks:~$ traceroute 10.50.0.10
traceroute to 10.50.0.10 (10.50.0.10), 30 hops max, 46 byte packets
 1  10.50.0.10 (10.50.0.10)  17.888 ms  19.571 ms  *
r-wks:~$ ping 10.50.0.10
PING 10.50.0.10 (10.50.0.10): 56 data bytes
64 bytes from 10.50.0.10: seq=0 ttl=42 time=13.743 ms
64 bytes from 10.50.0.10: seq=1 ttl=42 time=21.193 ms
64 bytes from 10.50.0.10: seq=2 ttl=42 time=14.960 ms
64 bytes from 10.50.0.10: seq=3 ttl=42 time=10.842 ms
64 bytes from 10.50.0.10: seq=4 ttl=42 time=15.058 ms
64 bytes from 10.50.0.10: seq=5 ttl=42 time=13.503 ms
^C
```

R-wks to DMZ-svcs



R-wks accessing the http

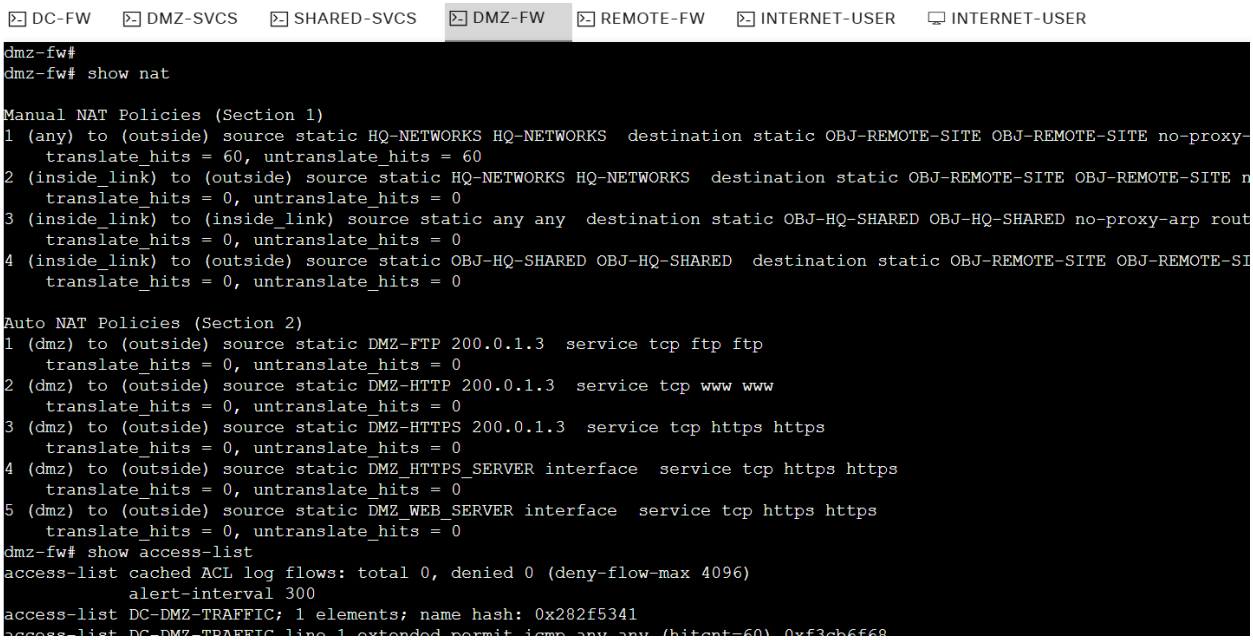
Task 7 – Publish DMZ Services for Internet Users

Objective

The objective of this task is to configure the firewall to securely publish internal DMZ services to internet users. Using Static Network Address Translation (Static NAT) with Port Redirection, the firewall maps specific public IP addresses and ports to internal DMZ servers that host services such as FTP, HTTP, and HTTPS.
Access to these services is controlled through Access Control Lists (ACLs) to ensure that only the required ports are allowed from the internet while maintaining the security of the internal network.
This configuration enables external users to access publicly available services hosted in the DMZ network while protecting the internal LAN infrastructure.

- **Verify NAT Configuration**

Command
#show nat



DC-FW | DMZ-SVCS | SHARED-SVCS | DMZ-FW | REMOTE-FW | INTERNET-USER | INTERNET-USER

```
access-list OUTSIDE_IN line 1 extended permit tcp any object DMZ-SVCS eq www (hitcnt=0) 0xa972d67b
  access-list OUTSIDE_IN line 1 extended permit tcp any host 10.50.0.10 eq www (hitcnt=0) 0xa972d67b
access-list OUTSIDE_IN line 2 extended permit tcp any object DMZ-SVCS eq https (hitcnt=0) 0x7f50a6a4
  access-list OUTSIDE_IN line 2 extended permit tcp any host 10.50.0.10 eq https (hitcnt=0) 0x7f50a6a4
access-list OUTSIDE_IN line 3 extended permit icmp any object DMZ-SVCS (hitcnt=0) 0x6cb2bdb0
  access-list OUTSIDE_IN line 3 extended permit icmp any host 10.50.0.10 (hitcnt=0) 0x6cb2bdb0
access-list OUTSIDE_IN line 4 extended permit tcp any host 10.50.0.10 eq ftp (hitcnt=0) 0x1605357a
access-list OUTSIDE_IN line 5 extended permit tcp any object DMZ_WEB_SERVER eq www (hitcnt=0) 0x811393da
  access-list OUTSIDE_IN line 5 extended permit tcp any host 10.50.0.10 eq www (hitcnt=0) 0x811393da
access-list OUTSIDE_IN line 6 extended permit tcp any object DMZ_WEB_SERVER eq https (hitcnt=0) 0xa98b914c
  access-list OUTSIDE_IN line 6 extended permit tcp any host 10.50.0.10 eq https (hitcnt=0) 0xa98b914c
access-list OUTSIDE_IN line 7 extended permit tcp any object DMZ_HTTPS_SERVER eq https (hitcnt=0) 0xbf37e791
  access-list OUTSIDE_IN line 7 extended permit tcp any host 10.50.0.10 eq https (hitcnt=0) 0xbf37e791
access-list OUTSIDE_IN line 8 extended deny icmp any object DMZ_HTTPS_SERVER (hitcnt=0) 0xd662b940
  access-list OUTSIDE_IN line 8 extended deny icmp any host 10.50.0.10 (hitcnt=0) 0xd662b940
access-list OUTSIDE_IN line 9 extended permit ip any any (hitcnt=1) 0x5bf8b50a
access-list OUTSIDE_IN line 10 remark --- ALLOW WEB ACCESS FOR INTERNET ---
access-list OUTSIDE_IN line 11 remark --- BLOCK PING FROM INTERNET ---
access-list OUTSIDE_IN line 12 extended deny icmp any any (hitcnt=0) 0xb487b3e1
access-list OUTSIDE_IN line 13 remark --- KEEP REMOTE-WORKSTATION/VPN ALIVE ---
access-list DMZ-OUT; 2 elements; name hash: 0x14523ada
access-list DMZ-OUT line 1 extended permit ip host 10.50.0.10 any (hitcnt=0) 0x33767c65
access-list DMZ-OUT line 2 extended permit icmp any any (hitcnt=0) 0x59010cae
access-list VPN-TRAFFIC; 4 elements; name hash: 0xb42654bc
access-list VPN-TRAFFIC line 1 extended permit ip object-group HQ-NETWORKS object OBJ-REMOTE-SITE (hitcnt=60) 0x7d36ec63
  access-list VPN-TRAFFIC line 1 extended permit ip 10.50.0.0 255.255.255.0 192.168.100.0 255.255.255.0 (hitcnt=60) 0x16345f43
  access-list VPN-TRAFFIC line 1 extended permit ip 10.1.10.0 255.255.255.0 192.168.100.0 255.255.255.0 (hitcnt=0) 0x15b68bae
  access-list VPN-TRAFFIC line 1 extended permit ip 192.168.10.0 255.255.255.0 192.168.100.0 255.255.255.0 (hitcnt=0) 0x3de884e4
  access-list VPN-TRAFFIC line 1 extended permit ip 192.168.20.0 255.255.255.0 192.168.100.0 255.255.255.0 (hitcnt=0) 0x08a153d6
```

DC-FW | DMZ-SVCS | SHARED-SVCS | DMZ-FW | REMOTE-FW | INTERNET-USER | INTERNET-USER

```
  access-list OUTSIDE-IN line 1 extended permit ip 192.168.100.0 255.255.255.0 10.50.0.0 255.255.255.0 (hitcnt=0) 0xb39ebf0a
  access-list OUTSIDE-IN line 1 extended permit ip 192.168.100.0 255.255.255.0 10.1.10.0 255.255.255.0 (hitcnt=0) 0x9b351016
  access-list OUTSIDE-IN line 1 extended permit ip 192.168.100.0 255.255.255.0 192.168.10.0 255.255.255.0 (hitcnt=0) 0x8afe1640
  access-list OUTSIDE-IN line 1 extended permit ip 192.168.100.0 255.255.255.0 192.168.20.0 255.255.255.0 (hitcnt=0) 0x1d83f30b
access-list OUTSIDE-IN line 2 extended permit udp host 200.0.2.2 host 200.0.1.2 eq isakmp (hitcnt=0) 0x9a0a8413
access-list OUTSIDE-IN line 3 extended permit esp host 200.0.2.2 host 200.0.1.2 (hitcnt=0) 0x866e02db
access-list OUTSIDE-IN line 4 extended permit tcp any host 200.0.1.3 eq www (hitcnt=0) 0xcabd76ae
access-list OUTSIDE-IN line 5 extended permit tcp any host 200.0.1.3 eq https (hitcnt=0) 0x6b67ee18
access-list OUTSIDE-IN line 6 extended permit tcp any host 10.50.0.10 eq www (hitcnt=0) 0x10efa68f
access-list OUTSIDE-IN line 7 extended permit tcp any host 10.50.0.10 eq https (hitcnt=0) 0xf85f9c87
access-list IT-IN; 1 elements; name hash: 0xf7e48bf5
access-list IT-IN line 1 extended permit ip 192.168.10.0 255.255.255.0 10.1.10.0 255.255.255.0 (hitcnt=0) 0xd6714c90
access-list INSIDE_IN; 1 elements; name hash: 0xf4313c68
access-list INSIDE_IN line 1 extended permit icmp any any (hitcnt=0) 0x0ef125bf
access-list DMZ_TO_INTERNAL; 3 elements; name hash: 0x8ac00291
access-list DMZ_TO_INTERNAL line 1 extended permit udp host 10.50.0.10 host 10.1.10.10 eq ntp (hitcnt=0) 0x33228448
access-list DMZ_TO_INTERNAL line 2 extended permit udp host 10.50.0.10 host 10.1.10.10 eq domain (hitcnt=0) 0xa3f9391c
access-list DMZ_TO_INTERNAL line 3 extended permit icmp host 10.50.0.10 host 10.1.10.10 (hitcnt=0) 0x552dfce3
access-list DC-INBOUND; 1 elements; name hash: 0xec2b2aa4
access-list DC-INBOUND line 1 extended permit icmp any any (hitcnt=0) 0x4776fd87
```

Figure: Static NAT configuration on the firewall mapping the DMZ server (10.50.0.10) to the public IP address (200.0.1.3) for FTP, HTTP, and HTTPS services.

- **ACL Verification**

Command

#show access-list

```
access-list OUTSIDE-IN line 4 extended permit tcp any host 200.0.1.3 eq www (hitcnt=0) 0xcabd76ae
access-list OUTSIDE-IN line 5 extended permit tcp any host 200.0.1.3 eq https (hitcnt=0) 0x6b67ee18
access-list OUTSIDE-IN line 6 extended permit tcp any host 10.50.0.10 eq www (hitcnt=0) 0x10efa68f
access-list OUTSIDE-IN line 7 extended permit tcp any host 10.50.0.10 eq https (hitcnt=0) 0xf85f9c87
access-list IT-IN; 1 elements; name hash: 0xf7e48bf5
access-list IT-IN line 1 extended permit ip 192.168.10.0 255.255.255.0 10.1.10.0 255.255.255.0 (hitcnt=0) 0xd6714c90
access-list INSIDE_IN; 1 elements; name hash: 0xf4313c68
access-list INSIDE_IN line 1 extended permit icmp any any (hitcnt=0) 0x0ef125bf
access-list DMZ_TO_INTERNAL; 3 elements; name hash: 0x8ac00291
access-list DMZ_TO_INTERNAL line 1 extended permit udp host 10.50.0.10 host 10.1.10.10 eq ntp (hitcnt=0) 0x33228448
access-list DMZ_TO_INTERNAL line 2 extended permit udp host 10.50.0.10 host 10.1.10.10 eq domain (hitcnt=0) 0xa3f9391c
access-list DMZ_TO_INTERNAL line 3 extended permit icmp host 10.50.0.10 host 10.1.10.10 (hitcnt=0) 0x552dfce3
access-list DC-INBOUND; 1 elements; name hash: 0xec2b2aa4
access-list DC-INBOUND line 1 extended permit icmp any any (hitcnt=0) 0x4776fd87
```

Fig : Access Control List (OUTSIDE-IN) allowing internet users to access DMZ services such as HTTP, HTTPS, and FTP hosted on the DMZ server.

- **Firewall Interface Verification**

Command

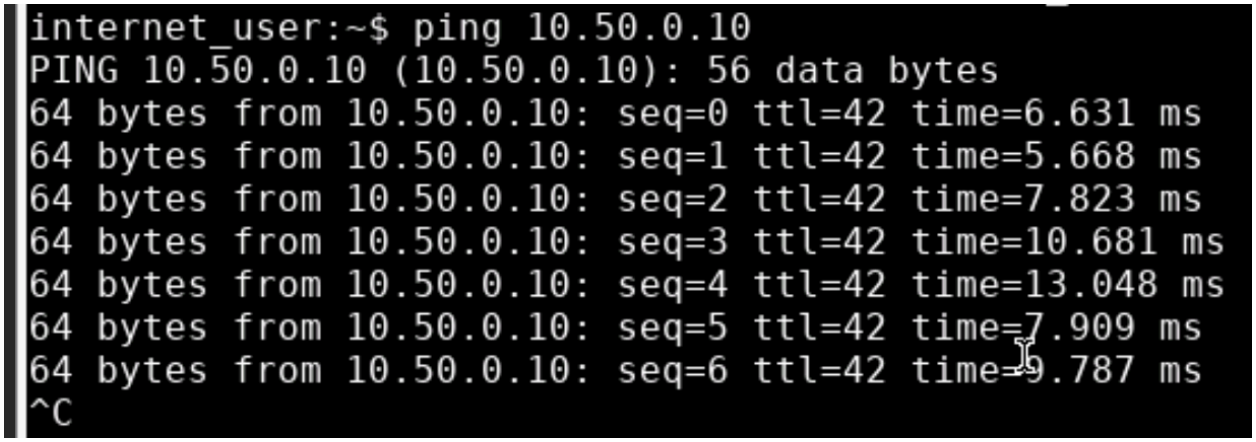
#show interface ip brief

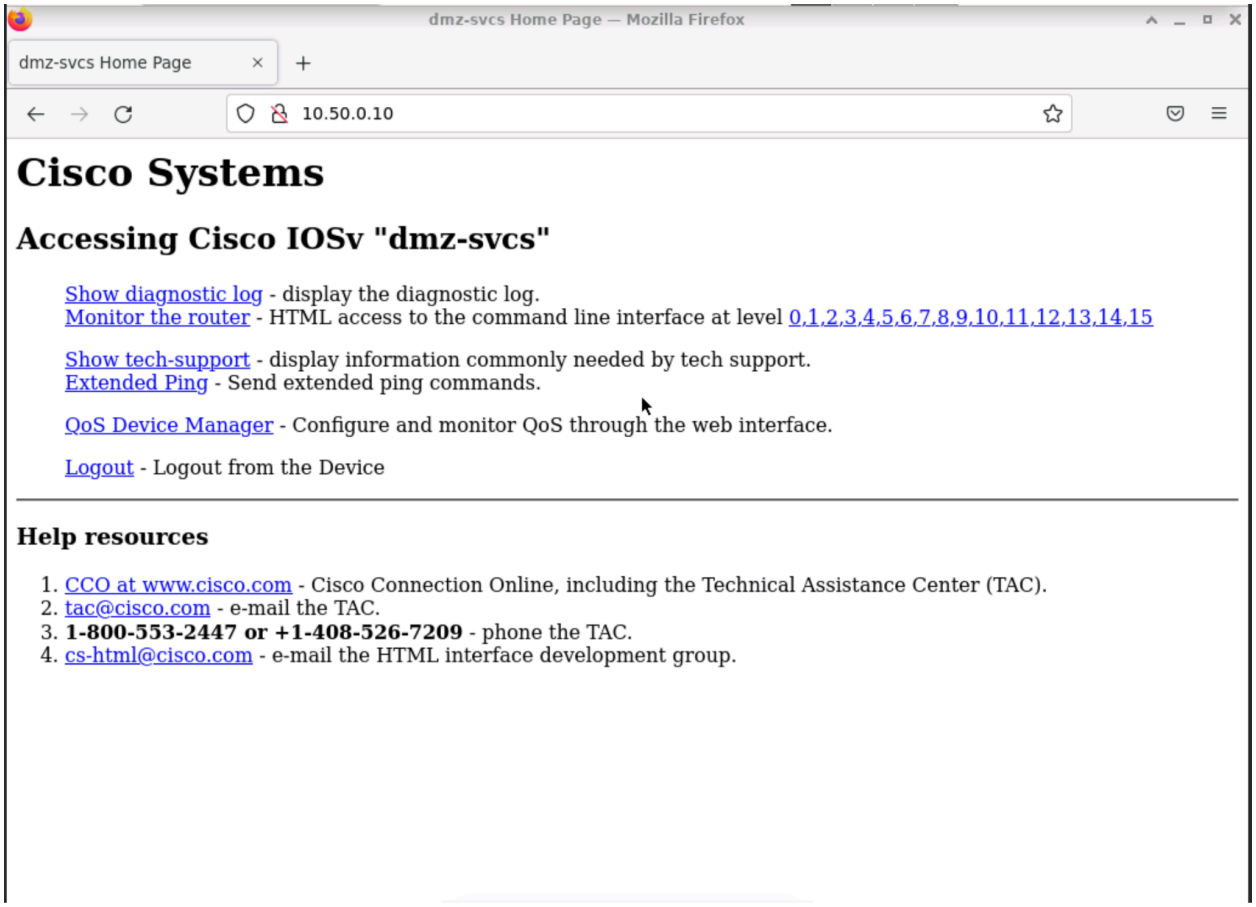
```
dmz-fw# show interface ip brief
Interface                  IP-Address      OK? Method Status                Protocol
GigabitEthernet0/0         10.0.0.26       YES CONFIG up                    up
GigabitEthernet0/1         10.50.0.1       YES CONFIG up                    up
GigabitEthernet0/2         200.0.1.2       YES CONFIG up                    up
Internal-Data0/0           169.254.1.1     YES unset  up                    up
Management0/0              unassigned      YES unset  administratively down up
dmz-fw#
```

Fig : Verification of firewall interface configuration showing inside, DMZ, and outside interfaces are operational.

```
    flags sIT idle 4:18:32 timeout 0:00:00
TCP PAT from dmz:10.50.0.10 21-21 to outside:200.0.1.3 21-21
    flags sr idle 4:18:32 timeout 0:00:00
TCP PAT from dmz:10.50.0.10 80-80 to outside:200.0.1.3 80-80
    flags sr idle 4:18:32 timeout 0:00:00
TCP PAT from dmz:10.50.0.10 443-443 to outside:200.0.1.3 443-443
    flags sr idle 4:18:35 timeout 0:00:00
TCP PAT from dmz:10.50.0.10 443-443 to outside:200.0.1.2 443-443
    flags sr idle 4:18:35 timeout 0:00:00
TCP PAT from dmz:10.50.0.10 443-443 to outside:200.0.1.2 443-443
    flags sr idle 4:18:35 timeout 0:00:00
```

Fig : NAT translation table showing port redirection between the public IP address (200.0.1.3) and the DMZ server (10.50.0.10).





Internet user

```
access-list OUTSIDE_IN line 1 extended permit tcp any object DMZ-SVCS eq www (hitcnt=12) 0xa972d67b
  access-list OUTSIDE_IN line 1 extended permit tcp any host 10.50.0.10 eq www (hitcnt=12) 0xa972d67b
access-list OUTSIDE_IN line 2 extended permit tcp any object DMZ-SVCS eq https (hitcnt=128) 0x7f50a6a4
  access-list OUTSIDE_IN line 2 extended permit tcp any host 10.50.0.10 eq https (hitcnt=128) 0x7f50a6a4
access-list OUTSIDE_IN line 3 extended permit icmp any object DMZ-SVCS (hitcnt=79) 0x6cb2bdb0
```

dmz-fw

TASK 8 Allow Internet Traffic from Internal VLANs and On-Demand Access for DMZ and DC Servers

Objective:

Students will configure the firewall to allow internet access for internal VLANs (IT-Admin and Business) while ensuring DMZ and DC servers to have an on-demand internet access.Additionally, they will advertise a default route into OSPF from the Internet-FW to provide a path for all internal networks to access the internet

To verify that the Internet-FW successfully advertises a "Gateway of Last Resort" (Default Route) to the internal topology using OSPF, ensuring all internal subnets can reach external networks.

Technical Evidence:

On the internal routing devices (e.g., dist-rtr), the routing table must be inspected for an externally learned OSPF route.

Command Executed: show ip route

Analysis of Results:

Upon execution, the routing table displays the following entry:

O*E2 0.0.0.0/0 [110/1] via 10.0.0.22, 00:24:12, GigabitEthernet0/1

Technical Breakdown:

- O* Symbol: Indicates a candidate default route learned via the OSPF protocol.
- E2 Designation: Specifies an External Type 2 route. This confirms the route was "redistributed" or "originated" from an external autonomous system (the Internet) into our OSPF domain by the Internet-FW.
- 0.0.0.0/0: The quad-zero notation represents the default route, covering all IP addresses not explicitly found in the routing table.
- via 10.0.0.22: Confirms that all internet-bound traffic is being correctly forwarded to the Internet-FW inside interface.

```
dist-rtr#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       a - application route
       + - replicated route, % - next hop override, p - overrides from PfR

Gateway of last resort is 10.0.0.13 to network 0.0.0.0

O*E2  0.0.0.0/0 [110/1] via 10.0.0.13, 3d20h, GigabitEthernet0/2
                [110/1] via 10.0.0.2, 3d19h, GigabitEthernet0/1
      10.0.0.0/8 is variably subnetted, 11 subnets, 3 masks
C        10.0.0.0/30 is directly connected, GigabitEthernet0/1
L        10.0.0.1/32 is directly connected, GigabitEthernet0/1
O        10.0.0.4/30 [110/2] via 10.0.0.2, 3d19h, GigabitEthernet0/1
O        10.0.0.8/30 [110/2] via 10.0.0.13, 3d20h, GigabitEthernet0/2
C        10.0.0.12/30 is directly connected, GigabitEthernet0/2
L        10.0.0.14/32 is directly connected, GigabitEthernet0/2
O IA     10.0.0.16/30 [110/2] via 10.0.0.2, 3d19h, GigabitEthernet0/1
O IA     10.0.0.20/30 [110/3] via 10.0.0.13, 3d20h, GigabitEthernet0/2
                      [110/3] via 10.0.0.2, 3d19h, GigabitEthernet0/1
O IA     10.0.0.24/30 [110/2] via 10.0.0.13, 3d20h, GigabitEthernet0/2
O IA     10.1.10.0/24 [110/12] via 10.0.0.2, 3d18h, GigabitEthernet0/1
O IA     10.50.0.0/24 [110/12] via 10.0.0.13, 3d20h, GigabitEthernet0/2
      192.168.10.0/24 is variably subnetted, 2 subnets, 2 masks
C        192.168.10.0/24 is directly connected, GigabitEthernet0/0.10
L        192.168.10.1/32 is directly connected, GigabitEthernet0/0.10
      192.168.20.0/24 is variably subnetted, 2 subnets, 2 masks
C        192.168.20.0/24 is directly connected, GigabitEthernet0/0.20
L        192.168.20.1/32 is directly connected, GigabitEthernet0/0.20
dist-rtr#
```

Dist-rtr

```
dmz-rtr#sh ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       a - application route
       + - replicated route, % - next hop override, p - overrides from PfR

Gateway of last resort is 10.0.0.9 to network 0.0.0.0

O*E2  0.0.0.0/0 [110/1] via 10.0.0.9, 3d20h, GigabitEthernet0/3
      10.0.0.0/8 is variably subnetted, 12 subnets, 3 masks
O        10.0.0.0/30 [110/2] via 10.0.0.14, 3d20h, GigabitEthernet0/2
```

Dmz-rtr

```
wan-rtr#sh ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       a - application route
       + - replicated route, % - next hop override, p - overrides from PfR

Gateway of last resort is 10.0.0.22 to network 0.0.0.0

O*E2  0.0.0.0/0 [110/1] via 10.0.0.22, 6d04h, GigabitEthernet0/0
      10.0.0.0/8 is variably subnetted, 12 subnets, 3 masks
O        10.0.0.0/30 [110/2] via 10.0.0.5, 3d18h, GigabitEthernet0/4
C        10.0.0.4/30 is directly connected, GigabitEthernet0/4
```

Wan-rtr

```
dc-rtr#sh ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       a - application route
       + - replicated route, % - next hop override, p - overrides from PfR

Gateway of last resort is 10.0.0.6 to network 0.0.0.0

O*E2  0.0.0.0/0 [110/1] via 10.0.0.6, 3d18h, GigabitEthernet0/4
      10.0.0.0/8 is variably subnetted, 12 subnets, 3 masks
C        10.0.0.0/30 is directly connected, GigabitEthernet0/1
```

Dc-rtr

Conclusion:

The presence of the O*E2 entry validates the successful execution of the default-information originate command on the edge firewall. The internal network is now "routing aware" of the exit path to the ISP, eliminating the need for manual static routing on every internal hop.

To verify that internal subnets (IT-Admin, Business, DMZ, and DC) are successfully translated to a single public IP address for internet access while maintaining a secure, stateful connection.

Technical Evidence of internet-fw:

The Internet-FW (ASA) was inspected using the show nat detail and show xlate commands to confirm that Private-to-Public translation is occurring.

Command Executed: show nat detail

Analysis of Results:

- NAT Section 2 (Auto NAT): The output shows a rule for INTERNAL-NETS translating to the interface (the public IP 200.0.0.2).
- Translate Hits: 79,521 hits. This high number is proof that thousands of internal packets have been successfully translated and sent to the internet.
- Untranslate Hits: These are the return packets coming back from the internet. The firewall correctly matched these to existing internal requests.

Technical Breakdown for the Report:

1. Dynamic PAT (Port Address Translation): By using the source dynamic any interface keyword, the firewall is performing PAT. This allows all internal users to share the single public IP address (200.0.0.2) by assigning unique source port numbers to each session.
2. Topology Hiding: The Origin: 0.0.0.0/0 to Translated: 200.0.0.2 mapping confirms that the internal private addressing scheme (10.x.x.x) is completely hidden from the public internet, fulfilling a key security requirement.
3. Stateful On-Demand Logic: The show conn command (showing 2 in use) confirms that the firewall is tracking "state." It only allows traffic back into the DMZ or DC servers if those servers initiated the request first. If an external attacker tried to reach the servers without a matching entry in the show conn table, the ASA would drop the packet.

```
internet-fw# show ospf database external


            OSPF Router with ID (10.0.0.22) (Process ID 6500)

                Type-5 AS External Link States

  LS age: 311
  Options: (No TOS-capability, DC)
  LS Type: AS External Link
  Link State ID: 0.0.0.0 (External Network Number )
  Advertising Router: 10.0.0.22
  LS Seq Number: 80000116
  Checksum: 0xd629
  Length: 36
  Network Mask:0.0.0.0
        Metric Type: 2 (Larger than any link state path)
        TOS: 0
        Metric: 1
        Forward Address: 0.0.0.0
        External Route Tag: 6500

internet-fw#
```

Internet-fw

```
internet-fw# show xlate
1 in use, 118 most used
Flags: D - DNS, e - extended, I - identity, i - dynamic, r - portmap,
       s - static, T - twice, N - net-to-net
NAT from outside:0.0.0.0/0 to inside:0.0.0.0/0
    flags sIT idle 125:39:32 timeout 0:00:00
```

```
internet-fw# show nat detail

Manual NAT Policies (Section 1)
1 (inside) to (outside) source dynamic any interface
    translate_hits = 3478, untranslate_hits = 17
    Source - Origin: 0.0.0.0/0, Translated: 200.0.0.2/24

Auto NAT Policies (Section 2)
1 (inside) to (outside) source dynamic INTERNAL-NETS interface
    translate_hits = 79521, untranslate_hits = 7
    Source - Origin: 0.0.0.0/0, Translated: 200.0.0.2/24
```

```
internet-fw# show conn
2 in use, 100 most used
```

internet-fw

Conclusion:

The NAT configuration is verified and fully operational. The high hit count on the dynamic NAT policy confirms that the OSPF default route is successfully driving traffic to the firewall, where it is then translated and forwarded to the ISP. This setup ensures seamless internet connectivity for users while strictly controlling outbound access for critical server infrastructure.

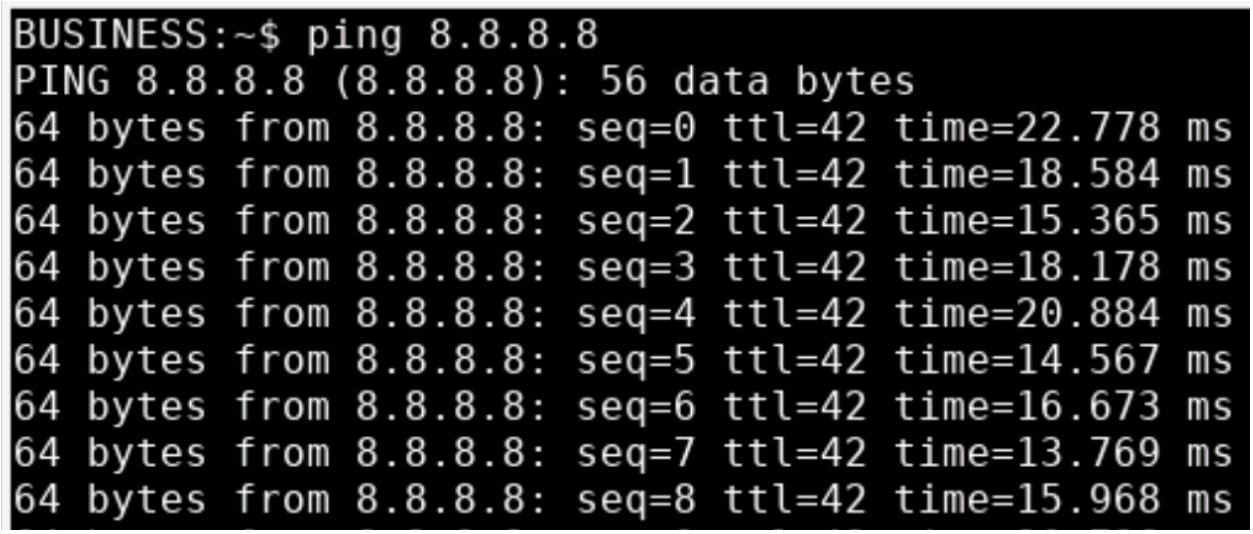
To demonstrate that internal users have "On-Demand" internet access while maintaining a "Deny-by-Default" posture for all inbound traffic from the ISP.

Test Case A: Internal to External (Successful)

- **Action:** Ping from Business PC (10.x.x.x) to ISP Gateway.
- **Result: Success.**
- **Significance:** Proves that OSPF routing, Dynamic NAT (PAT), and Outbound ACLs are correctly configured to allow "On-Demand" internet requests.

Test Case B: External to Internal (Failed)

- **Action:** Ping from ISP Gateway to Business PC Private IP.
- **Result: Failed (Request Timed Out).**
- **Significance:** Proves the Firewall is performing **Stateful Inspection**. It protects the internal network by rejecting unsolicited traffic and hiding the internal topology from the public domain.



Business-to ISP

```
ISP#ping 192.168.20.3
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.20.3, timeout is 2 seconds:
.....
Success rate is 0 percent (0/5)
```

Isp to business it fails

Task 9 – Centralized Access Through Shared-Services Server

Objective

The objective of this task is to implement **centralized administrative access** to all network devices within the enterprise network. Instead of allowing administrators to connect to routers, switches, and firewalls from multiple hosts, the design enforces that **all management access must originate from the Shared-Services server located in the Data Center network**.

This approach improves security by reducing the attack surface and ensuring that administrative activities are performed through a controlled management host.

Security Mechanisms Implemented

To enforce centralized management, the following mechanisms were implemented on all Cisco IOS devices.

1. Local User Authentication

Each device uses **locally configured administrative credentials**.

Example configuration:

#username admin privilege 15 secret cisco

This ensures that only authenticated administrators can access the devices.

2. Secure Remote Management Using SSH

SSH version 2 was enabled to provide **encrypted management access**.

Example configuration:

#ip domain-name cisco.lab
#crypto key generate rsa modulus 1024
#ip ssh version 2

SSH protects administrative credentials and management traffic from interception.

3. Restricting Management Access with Access Control Lists

A dedicated ACL named **MGMT_FROM_SHARED** was created to allow management access **only from the Shared-Services server (10.1.10.10)**.

Example ACL configuration:

```
#ip access-list standard MGMT_FROM_SHARED
#permit 10.1.10.10
#deny any
```

This prevents other hosts in the network from directly managing infrastructure devices.

4. Applying Access Control to VTY Lines

The ACL was applied to the VTY lines to enforce centralized management.

Example configuration:

```
line vty 0 4
 login local
 transport input ssh
 access-class MGMT_FROM_SHARED in
```

This configuration ensures that:

- Only **SSH** connections are permitted
- Only **Shared-Services server** can initiate the connection
- All administrators must **authenticate locally**
- **Configuration Verification**

After configuration, several commands were used to verify the centralized access implementation.

Verification 1 – Local User Configuration

Command:

show run | sec username

```
dist-rtr#
dist-rtr#show run | sec username
username admin privilege 15 secret 9 $9$LW62WeLIzQ5Gp9$DVUJu3wSQ6dyPB9L.X53hs39A0/XSmEZNM15Xbf8BkQ
dist-rtr#
```

Fig : Local administrative user configured on the router for authentication.

Verification 2 – VTY Line Security Configuration

Command:

show run | sec line vty

```
dist-rtr#
dist-rtr#show run | sec line vty
line vty 0 4
 access-class MGMT_FROM_SHARED in
 password cisco
 login local
 transport input ssh
dist-rtr#
```

Fig : VTY line configuration restricting remote management to SSH and requiring local authentication.

Verification 3 – Management Access Control List

Command:

show access-lists

```
 transport input ssh
dist-rtr#show access-lists
Standard IP access list MGMT_FROM_SHARED
    10 permit 10.1.10.10 (4 matches)
    20 deny   any
dist-rtr#
```

ACL restricting administrative access to the Shared-Services server.

Verification 4 – SSH Status

Command:

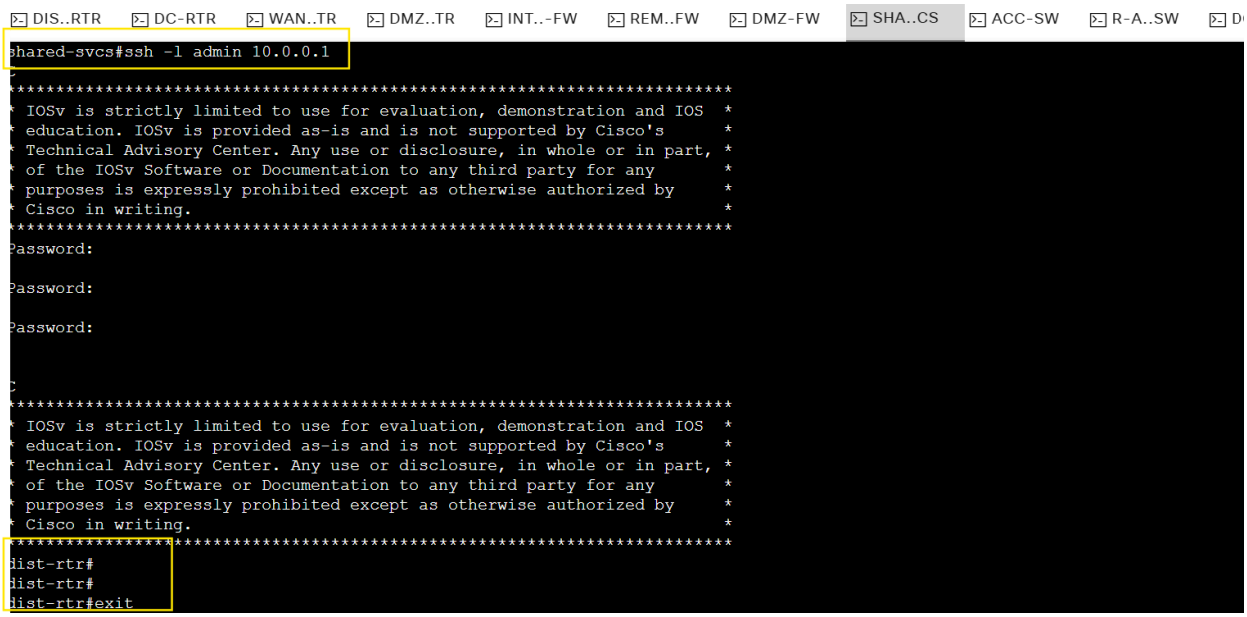
show ip ssh

```
dist-rtr#show ip ssh
SSH Enabled - version 2.0
Authentication methods:publickey,keyboard-interactive,password
Authentication Publickey Algorithms:x509v3-ssh-rsa,ssh-rsa
Hostkey Algorithms:x509v3-ssh-rsa,ssh-rsa
Encryption Algorithms:aes128-ctr,aes192-ctr,aes256-ctr
MAC Algorithms:hmac-sha2-256,hmac-sha2-512,hmac-sha1,hmac-sha1-96
KEX Algorithms:diffie-hellman-group-exchange-sha1,diffie-hellman-group14-sha1
Authentication timeout: 120 secs; Authentication retries: 3
Minimum expected Diffie Hellman key size : 2048 bits
IOS Keys in SECSH format(ssh-rsa, base64 encoded): dist-rtr.cisco.lab
ssh-rsa AAAAB3NzaC1yc2EAAAADAQABAAAAgQCr632Z1YPecrWhi/O+XztoIBrcSNQgwJ6p7Lgwp7EY
28df4ubzMU8bLba8agCDErQ0xyrIXvX7DVHkffJs3/FYQSl6Z2RP2bxY0XDLRDLQud0ZuPiv1scqrwMI
4vPPxC2LwCY6cLG80JCSjAHZmzkpS9gt7nWEAQju0fAFmLdT6Q==
dist-rtr#
```

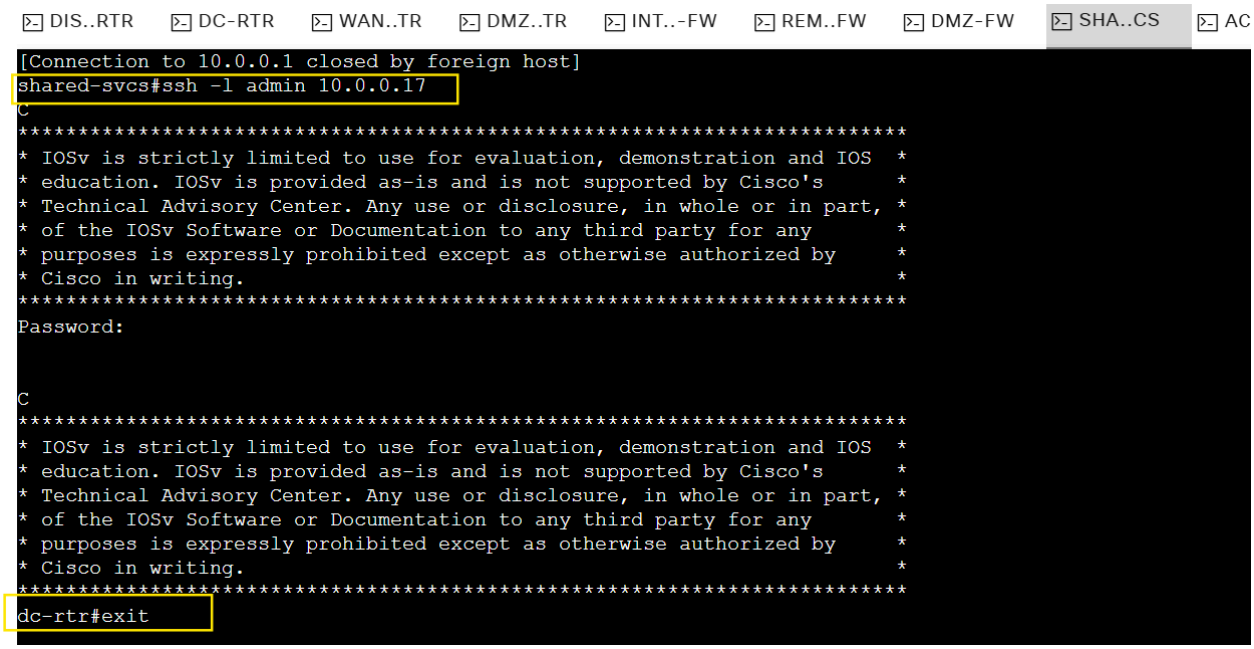
Fig : SSH service enabled on the router for secure remote management.

Verification 5 – Successful SSH Access

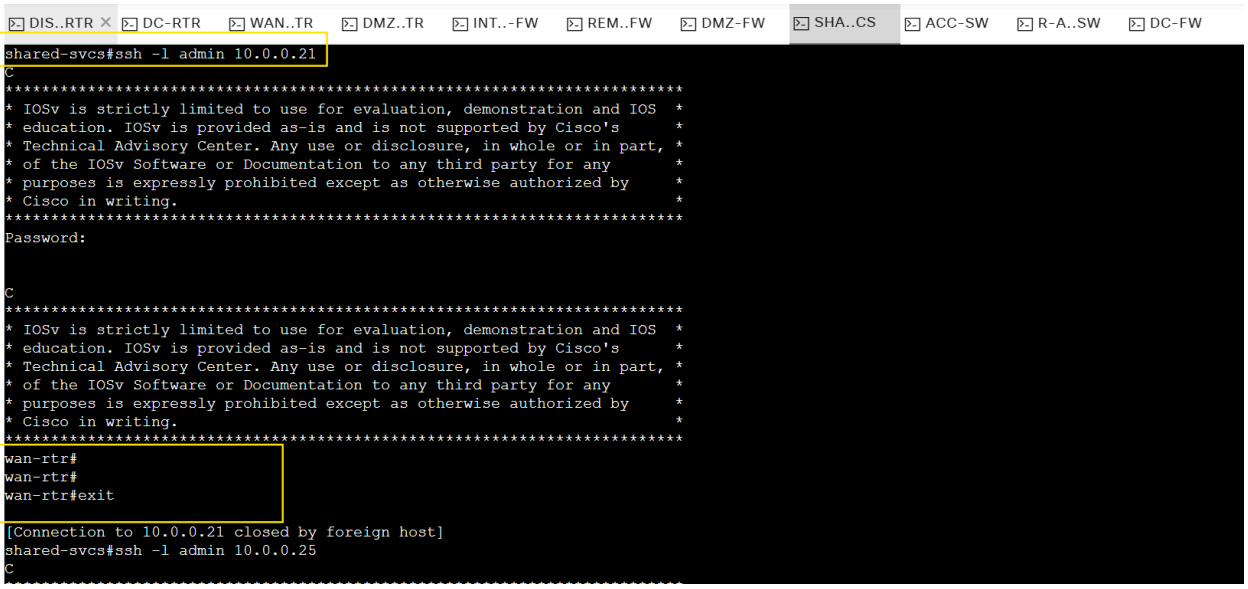
From the Shared-Services server:



Successful SSH login from the Shared-Services server to a network router.



Successful SSH login from the Shared-Services server to a network router.



Successful SSH login from the Shared-Services server to a network router.



Successful SSH login from the Shared-Services server to a network router.

Conclusion

In this task, centralized management of network infrastructure was successfully implemented using the Shared-Services server. All routers, switches, and firewalls were configured to allow administrative access exclusively through this server using secure SSH connections and locally authenticated users.

Access control lists were applied to restrict management traffic to the server's IP address, ensuring that no other hosts can directly manage the network devices. This architecture enhances the overall security of the enterprise network by enforcing controlled, authenticated, and encrypted administrative access.

Task 10 – Implementation of Layer 2 Security Features

Objective

The objective of this task is to enhance the security of the enterprise network by implementing key **Layer 2 security mechanisms** on the access switches. These mechanisms protect the network from common internal attacks such as:

- MAC address spoofing
- Rogue DHCP servers
- ARP poisoning attacks

To achieve this objective, the following Layer 2 security features were implemented on the access switches:

1. **Port Security**
2. **DHCP Snooping**
3. **Dynamic ARP Inspection (DAI)**

These features work together to provide a secure and controlled Layer 2 environment.

NOTE: Network Behavior After Applying Security Features

After applying the Layer-2 security configurations, it was observed that the entire network temporarily experienced connectivity issues. Several devices lost network access and communication between hosts was disrupted.
This behavior occurred because the newly enabled security mechanisms immediately started enforcing strict validation rules.
Specifically:

- **Port Security** shut down interfaces when MAC address violations were detected.
- **DHCP Snooping** blocked DHCP messages from interfaces that were not marked as trusted.
- **Dynamic ARP Inspection** dropped ARP packets that did not match entries in the DHCP snooping binding table.

Since the network was previously operating without these strict security checks, the activation of these mechanisms caused several ports and services to be temporarily restricted.

So we remove some services and attached the configuration here.

Port Security Configuration

Port Security was configured on access ports connected to end devices to prevent unauthorized hosts from connecting to the network.

Configuration

```
#conf t
#interface range GigabitEthernet0/1 - 3
 #switchport mode access
 #switchport port-security
 #switchport port-security maximum 1
 #switchport port-security violation shutdown
 #switchport port-security mac-address sticky
#end
#wr
```

Explanation

Command	Description
switchport port-security	Enables port security on the interface
maximum 1	Allows only one MAC address per port
violation shutdown	Shuts down the port if violation occurs
mac-address sticky	Automatically learns and stores the MAC address

This ensures that only authorized devices can connect to the switch ports.

DHCP Snooping Configuration

DHCP Snooping was configured to prevent unauthorized DHCP servers from assigning IP addresses to hosts in the network.

Configuration

```
#conf t

#ip dhcp snooping
#ip dhcp snooping vlan 10
#ip dhcp snooping vlan 20

#interface GigabitEthernet0/0
 #ip dhcp snooping trust
```

Explanation

Command	Description
ip dhcp snooping	Enables DHCP snooping globally
ip dhcp snooping vlan	Enables DHCP snooping for specific VLANs
ip dhcp snooping trust	Marks the interface connected to the legitimate DHCP server

Ports connected to end devices remain **untrusted by default**, preventing rogue DHCP servers.

Dynamic ARP Inspection (DAI)

Dynamic ARP Inspection was implemented to prevent ARP spoofing attacks by validating ARP packets using the DHCP snooping binding table.

Configuration

```
conf t

ip arp inspection vlan 10
ip arp inspection vlan 20

interface GigabitEthernet0/0
 ip arp inspection trust

end
wr
```

Explanation

Command	Description
ip arp inspection vlan	Enables ARP inspection on specific VLANs
ip arp inspection trust	Allows ARP packets on trusted interfaces

All other ports are treated as untrusted and ARP packets are validated before forwarding.

Conclusion

In this task, essential Layer-2 security mechanisms were implemented to protect the enterprise network from common internal attacks. Port Security was configured to restrict unauthorized devices, DHCP Snooping was implemented to prevent rogue DHCP servers, and Dynamic ARP Inspection was enabled to protect against ARP spoofing attacks.

Although enabling these security mechanisms initially caused network disruptions, this behavior demonstrated the strict enforcement of security policies and highlighted the importance of carefully configuring trusted interfaces in secure network environments.

CONCLUSION

This project successfully demonstrates the implementation of a high-security, multi-site enterprise network that balances seamless connectivity with rigorous protection. We built a hardened routing core where **OSPF** is secured with **MD5 authentication**, ensuring that the network map cannot be tampered with by unauthorized devices. User access was streamlined through **DHCP** delivery using IP-helper logic on the distribution layer, while our public-facing services on the **DMZ-SVCS** were secured by supporting both **HTTP** and encrypted **HTTPS** traffic. On the management plane, we enforced a strict security policy by transitioning from insecure, clear-text **Telnet** to **SSH v2 with RSA keys**. This ensures that administrative access to the Core and Firewalls is fully encrypted and protected against credential sniffing. To keep the configuration clean and scalable, we utilized **Network Objects** on the ASA firewalls. This object-oriented approach allowed us to write precise Access Control Lists (ACLs) that are easy to manage and audit. The final layer of defense was the **Internet-FW** and **Site-to-Site VPN** configuration. By implementing **Dynamic NAT (PAT)**, we provided "On-Demand" internet access to the Business and IT-Admin VLANs while keeping our internal 10.x.x.x topology completely invisible to the ISP. This **Topology Hiding**, combined with the encrypted **IPsec tunnel** for remote office communication and **NTP** for global time synchronization, results in a professional-grade infrastructure where every service is optimized for security, visibility, and resilience.